Universal Quantum Perceptrons for Quantum Machine Learning

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Quantum neuromorphic computing (QNC) is a subfield of quantum machine learning (QML) that capitalizes on inherent system dynamics. As a result, QNC can run on contemporary, noisy quantum hardware and is poised to realize challenging algorithms in the near term. One key element yet to be added to QNC is the characterization of the requisite dynamics for universal quantum neuromorphic computation. We address this issue by proposing a quantum equivalent to the classical perceptron, a simple mathematical model for a neuron that is the building block of various machine learning architectures. We introduce a quantum perceptron (QP) based on the analog dynamics of interacting qubits with tunable coupling constants. By adding tunable single-qubit rotations to the QP, we demonstrate that a QP can realize universal quantum computation, which contrasts sharply with the limited computational complexity of a single classical perceptron. We show that QPs are analogous to variational quantum algorithms (VQAs) familiar to the quantum machine learning community. We derive the quantum neural tangent kernel of a QP and compare the QP's trainability to the trainability of other VQAs. We discuss the advantages and drawbacks of kernel formalism. Finally, we demonstrate the effectiveness of QPs by applying them to numerous QML problems, including calculating the inner products between quantum states, entanglement witnessing, and quantum metrology.

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I. INTRODUCTION

Classical machine learning often employs simple mathematical models for neuron dynamics as computational building blocks [1]. Recent work in quantum machine learning (QML) algorithm design illuminates the similarities between the dynamics of neurons in the brain and those of noisy qubits [2, 3]. These studies, named quantum neuromorphic computing (QNC) [4, 5], promise to deliver noise-resilient, scalable algorithms for QML implementations in the near-term. An outstanding question in QNC is finding simple, scalable models for neuron dynamics which serve as the building blocks of quantum neuromorphic computing.

Indeed, QNC using superconducting qubits has already produced quantum analogs of memory units called quantum memory resistors (or quantum memristors) [6]. These quantum memristors realize quantum versions of realistic neuronal models such as the Hodgkin-Huxley model [7, 8]. However, this model is too complex to serve as a scalable building block. A more straightforward approach aims to quantize a mathematical model inspired by single neuron dynamics such as the perceptron [9]. Currently, there are several proposed quantum perceptrons, but they rely on computational architectures that are inaccessible in the near-term[10, 11], adiabatic computation [12, 13], or collective interaction [14]. Thus far, the perceptron has remained intractable on quantum hardware despite originating as a simple model for neuron dynamics and being recognized as an important open question in QML [15].

The present manuscript addresses this gap by proposing a novel version of a quantum perceptron (QP) based on the natural unitary dynamics of qubit ensembles. These dynamics, when supplemented with intermittent single-qubit rotations, are sufficient to recover the classical perceptron and enable universal quantum computation. On the other hand, a single perceptron cannot approximate generic functions. Classical perceptrons are universal approximators only when many perceptrons are stacked together [1]. Thus, a single QP's computational power outshines that of the classical perceptron at the expense of single-qubit rotations.

The structure of this manuscript is as follows. In Sec. II, we define QPs via both the unitary dynamics and the nonlinear activation function achieved by this architecture. However, the realized activation function in our QP differs from the classical activation functions typically used in classical perceptrons. In Sec. II C, we address the discrepancies between the activation functions and remedy them by equipping the QP with intermittent single-qubit rotations. These rotations suffice to approximate standard activation functions in ML in a noise-resilient way. Therefore, architectures built out of QPs promise to be noise-resilient. In Sec. III, we show that single-qubit rotations enable QPs to carry out universal quantum computation, thus proving the greater computational complexity of QPs relative to clas-

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sical perceptron. Sec. IV, we recast QPs in the language of variational quantum circuits (VQAs) as a quantum circuit composed of many-body unitary evolution and single-qubit rotations. We analytically derive QPs' learning theory using the quantum neural tangent kernel formalism — the ability of QPs to learn specific tasks in the large-depth limit. We provide numerical simulations to support our findings and discuss the advantages and drawbacks of this formalism. In Sec. V, we apply QPs to three important problems in quantum computing and QML: measuring quantum state inner products, entanglement detection via energy measurement, and quantum metrology via time-reversal operations. The strong performance of QPs on these tasks establishes that their dynamics enable learning. Sec. VII concludes by proposing possible hardware implementations of QPs and discussing future directions.

II. QUANTUM PERCEPTRONS

A. Review of classical perceptrons

In machine learning, the perceptron (also called the McCulloch-Pitts neuron) is a function that decides if an input belongs to a specific class [9]. Suppose we have an input represented by a vector $\boldsymbol{s} = (s_1, ..., s_N)$ where $s_i = \pm 1$. We define the perceptron as the mathematical operation

$$\boldsymbol{s} \to y(\boldsymbol{s}) = f\left(\sum_{i=1}^{N} J_i s_i - \Delta_o\right) \tag{1}$$

where J_i are weights for the different features s_i, Δ_o a bias, and f is a nonlinear function. The output is the single scalar $y(\boldsymbol{s})$. Typically, f is chosen to be the sigmoid function $f(x) = 1/(1 + e^{-x})$ or the rectilinear (ReLu) function where $f(x) = 0$ if $x < 0$ and otherwise $f(x) = x$. Fig. 1 shows a schematic of the perceptron (left).

In general, we seek weights and bias such that the perceptron approximates the desired operation $\tilde{y}(\boldsymbol{s})$ for N_s samples in a training set $\{\boldsymbol{s}_i\}_{i=1}^{N_s}$. To do this, one computes the square-loss

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{N_s}\sum_{i=1}^{N_s}(\tilde{y}(\boldsymbol{s}_i) - y(\boldsymbol{s}_i))^2, \tag{2}$$

where $\boldsymbol{\theta}$ is a vector made out of J_i and Δ_o. Then, we update the weights and bias via gradient descent

$$\boldsymbol{\theta}_i \to \boldsymbol{\theta}_i - \eta \frac{\partial \mathcal{L}}{\partial \boldsymbol{\theta}_i} \tag{3}$$

where η is the learning rate. The training set is separable if a hyperplane separates positive ($\tilde{y}(\boldsymbol{s}) > 0$) and negative ($\tilde{y}(\boldsymbol{s}) < 0$) inputs. For such a set, there exist weights J_i and bias Δ_o such that the perceptron's output converges to the correct classification, i.e., $y(\boldsymbol{s}) = \tilde{y}(\boldsymbol{s})$ [16]. However, if the training set is not separable in N dimensions, such convergence is not guaranteed.

One of the primary goals of machine learning is producing mathematical operations that can approximate any $\tilde{y}(\boldsymbol{s})$ to arbitrary precision. If an operation can approximate any function, it is known as a universal approximator. While a single perceptron is not a universal approximator, a neural network consisting of two layers is universal as long as the first layer contains a large number of perceptrons [1]. Fig. 1 shows a schematic of this multilayer-perceptron at the bottom left.

B. Quantization of the perceptron

We define a quantum perceptron (QP) as a set of $N+1$ qubits with N inputs nodes labeled by $i = 1, ..., N$ and a single output node labeled by o. Each qubit inhabits the Hilbert space $\mathcal{H}_n$ spanned by the basis vectors $\{|0\rangle_n, |1\rangle_n\}$ such that the entire system is in the product Hilbert space $\mathcal{H} = \bigotimes_{n=1}^{N+1} \mathcal{H}_n$. The system evolves under the following Hamiltonian

$$H_P = \hat{h}\sigma_o^z + \Omega_o \sigma_o^x \tag{4}$$

$$\hat{h} = -\Delta_o + \sum_{i=1}^{N} J_i \sigma_i^z. \tag{5}$$

where $\sigma_n^z = |0\rangle\langle 0|_n - |1\rangle\langle 1|_n$ is the Pauli-Z operator and $\sigma_n^x = |0\rangle\langle 1|_n + |1\rangle\langle 0|_n$ is the Pauli-X operator. The parameter Δ_o is a field on the output qubit corresponding to an energy mismatch between the field's frequency and the transition frequency of the qubit. The field Ω_o is a coherent drive coupling the states $|0\rangle_o$ and $|1\rangle_o$ on the output qubit. The parameters $\{J_i\}_i$ are the coupling strengths between the input and output qubits. We highlight that H_P is related to the central-spin model describing, for example, the interactions between a nitrogen-vacancy center in diamond and neighboring nitrogen impurities [17]. Here, we assume the system evolves for a total time $\tau = 1$. We follow Schrödinger's equation: the initial state $|\Psi_0\rangle$ evolves to $e^{iH_P}|\Psi_0\rangle$.

Along this line of thinking, let us initialize the system in a state $|\Psi_0\rangle = \sum_{\boldsymbol{s}} \psi_{\boldsymbol{s}} |\boldsymbol{s}\rangle |0\rangle_o$ where $|\boldsymbol{s}\rangle$ is an eigenstate for each Pauli-Z on the input (i.e., $\sigma_i^z |\boldsymbol{s}\rangle = s_i |\boldsymbol{s}\rangle$ with $s_i = \pm 1$). According to Eq. (4) (see Appx. A for details) the system evolves to

$$\sum_{\boldsymbol{s}} \psi_{\boldsymbol{s}} |\boldsymbol{s}\rangle \left[\cos\omega(\boldsymbol{s}) - i\frac{h(\boldsymbol{s})}{\omega(\boldsymbol{s})}\sin\omega(\boldsymbol{s})\right]|0\rangle_o + \sum_{\boldsymbol{s}} \psi_{\boldsymbol{s}} |\boldsymbol{s}\rangle\, i\frac{\Omega_o}{\omega(\boldsymbol{s})}\sin\omega(\boldsymbol{s})\,|1\rangle_o \tag{6}$$

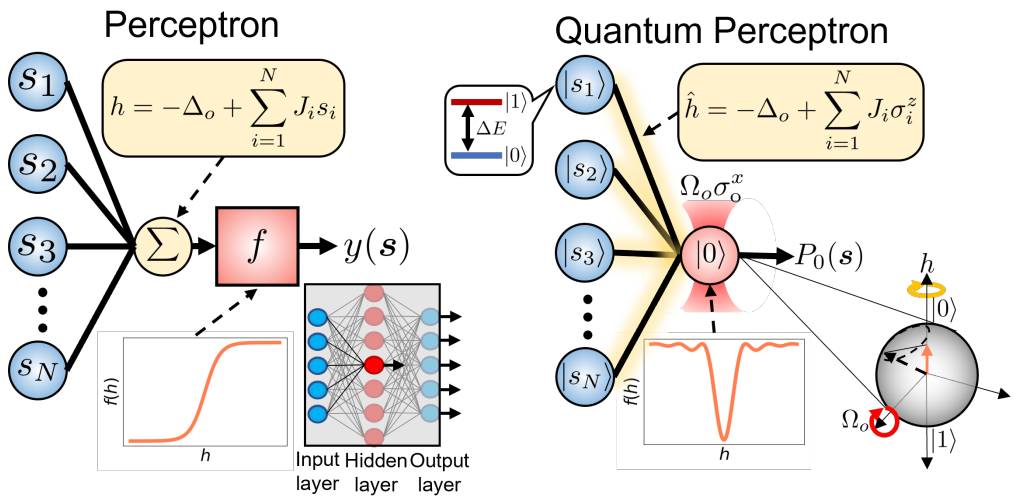

FIG. 1. Schematic of perceptrons. The classical perceptron (left) operates on the input N-dimensional vector $\boldsymbol{s}$ to produce an output $y(\boldsymbol{s}) = f(-\Delta_o + \sum_{i=1}^{N} J_i s_i)$ where J_i are weights and Δ_o a bias. The nonlinear function f is called the activation function. Here we exemplify f as the sigmoid function $f(h) = 1/(1+e^{-h})$. The weights and biases are chosen such that $y(\boldsymbol{s})$ approximates a target function $\tilde{y}(\boldsymbol{s})$, but a single perceptron cannot approximate all functions. However, as illustrated in the bottom left, neural networks composed of many perceptrons can approximate any function as long as the intermediate layer is sufficiently large. The proposed quantum perceptron (right) consists of N input qubits and one output qubit. The qubits evolve under the Hamiltonian in Eq. (4), which ensures that the probability $P_0(\boldsymbol{s})$ of the output qubit being in the state $|0\rangle_o$ is a nonlinear function of the input as in Eq. (6). We take $P_0(\boldsymbol{s})$ as the quantum perceptron's output. The inset on the bottom right shows the shape of the output for $\Omega_o = \pi$. At the bottom right, we illustrate the evolution of the output qubit. The output experiences a field Ω_o along the x-direction. The input qubits effectuate a field in the z-direction. The output qubit processes around the Bloch sphere due to these two fields. The state projection along $|0\rangle_o$ serves as the output.

where

$$h(\boldsymbol{s}) = \sum_{i=1}^{N} J_i s_i - \Delta_o, \tag{7}$$

$$\omega(\boldsymbol{s}) = \sqrt{h(\boldsymbol{s})^2 + \Omega_o^2}. \tag{8}$$

For convenience, we can recast the state above into the equation

$$\sum_{\boldsymbol{s}} \psi_{\boldsymbol{s}} |\boldsymbol{s}\rangle \left[\cos f(h(\boldsymbol{s})) |0\rangle_o + i \sin f(h(\boldsymbol{s})) |1\rangle_o\right]. \tag{9}$$

We call f the activation function inherent to Eq. (4), which is a nonlinear function of the input $\boldsymbol{s}$, and the parameters $\{J_i\}_i$, Δ_o, and Ω_o (see Eq. (A1) in Appx. A for the description of f). Fig. 1 (right) shows a schematic of the QP where the input qubits produce a field $\hat{h}$ on the output along the z-axis in addition to the Rabi drive Ω_o along the x-axis. We choose the probability of the output qubit being on state $|0\rangle_o$ after the evolution as the QP output. The bottom right inset shows the output function $f(h)$ for $\Omega_o = \pi$. Note that the output is a nonlinear function of h (and, thus, of the input).

Note that this activation function f is not the normal activation function encountered in machine learning applications but rather an oscillatory function on $h(\boldsymbol{s})$. In Sec. V A, we show that f helps calculate the inner product between quantum states, a central QML task. In Sec. II C, we explore how to modify f to recover canonical classical machine learning activation functions. Furthermore, recovering this classical limit provides insight from which we ultimately derive the increased computational power of QPs (see Sec. III).

C. Implementing activation functions via single-qubit rotations

We now focus on implementing the activation functions familiar to the machine learning community, which we refer to as f. The sigmoidal function $f(h) = 1/(1+e^{-h})$ is a popular example [18].

First of all, notice that we obtain $|0\rangle_o \to \cos f(h) |0\rangle_o - i \sin f(h) |1\rangle_o$ by evolving the output qubit under the unitary $U_f = e^{-if(h)\sigma_o^x}$. However, at first glance, the corresponding Hamiltonian is incommensurate with that of Eq. (4). Thus, we have to approximate f. To do this, we will assume that the activation function we desire has a Fourier series valid on an interval $[-\pi, \pi]$

$$f(h) = \frac{a_0}{2} + \sum_{k=1}^{\infty} \left(a_k \cos(kh) + b_k \sin(kh)\right) \tag{10}$$

with Fourier coefficients a_o, a_k, and b_k. In Appx. A, we show that

$$U_f = e^{-ia_0\sigma_N^x} \Pi_k W_k(h) V_k(h) \tag{11}$$

approximates f if

$$W_k(h) = \exp\left\{-i\frac{a_k}{2}\left(e^{-i\frac{kh}{2}\sigma_o^z}\sigma_o^x e^{i\frac{kh}{2}\sigma_o^z} + \text{h.c}\right)\right\}$$
$$V_k(h) = \exp\left\{i\frac{b_k}{2}\left(e^{-i(\frac{kh}{2}+\frac{\pi}{4})\sigma_o^z}\sigma_o^x e^{i(\frac{kh}{2}+\frac{\pi}{4})\sigma_o^z} + \text{h.c}\right)\right\}.$$

We realize the terms in the exponentials as follows. We let the output evolve under σ_o^x for a time proportional to a_k or b_k followed by intermittent rotations along the z-axis. We achieve the z-rotations by evolving the output under $h\sigma_o^z$ for specified amounts of time, namely $\tau = k$. These two steps make the result of Eq. (11) achievable via the Hamiltonian in Eq. (4) in a data-independent fashion.

This calculation illustrates that if we take Eq. (4) to have time-dependent fields or, equivalently, introduce digital control on the output qubit, we can generate any bounded and piece-wise continuous activation function. In the case of quantum perceptrons, the introduction of digital control amounts to rotating the output qubit. We note that, recently, Ref. [19] introduced a similar approach to approximating a single-variable function with applications to quantum neural networks.

In practice, we approximate the gate U_f by a finite number of rotations κ. Fig. 2 shows the results of approximating the activation functions $x^2, |x|$, sigmoid, and ReLu using $\kappa = 2, 5, 15$ Fourier coefficients. The largest discrepancies with the activation functions are confined to the edges of the interval on which the Fourier series is defined. In practice, there might be errors in each evolution's duration, generating additional discrepancies from the desired activation function. The inset in Fig. 2 shows the activation function when a_k, b_k, and h are added zero-mean Gaussian noise with standard deviation $\sigma = 0.01$. We note that the duration of the pulses of the field Ω_o correspond to the Fourier coefficients a_k and b_k. On the other hand, the duration of the pulses of the field h corresponds to the frequencies of the Fourier series. In Appx. A, we show that QPs are robust to noise in the field h. This result is due to the unitaries W_k and V_k, where h appears in conjugated operators. However, the perceptron is more sensitive to noise in the field Ω_o since this corresponds to changing the Fourier coefficients, and thus the function f is less well approximated.

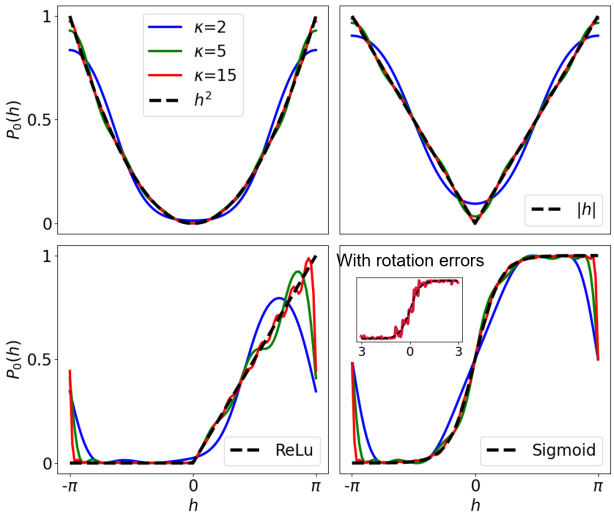

FIG. 2. Nonlinear functions approximated by a QP with single-qubit rotations on the output. For these approximations, we used the first 2 (blue), 5 (green), and 15 (red) Fourier coefficients of f. We plot the target functions for guidance (black dotted curves). The inset shows the approximated function on the interval $[-3, 3]$ with errors in both the values of the Fourier coefficients (x-rotations) and the value of the field h (z-rotations) for $f(h) = 1/(1 + e^{-h})$. In Appx. A, we show that QPs are robust to noise in the field h. This result is due to the unitaries W_k and V_k in Eq. (11), where h appears in conjugated operators.

Ref. [19] suggests approximating multi-variable functions by a multi-qubit system connected by CNOT gates. Our result in Eq. (11) makes this suggestion rigorous. From Eq. (11), it follows that a QP can implement any activation function with a Fourier series since we can approximate $f(\boldsymbol{s}) = f\left(\sum_i J_i s_i - \Delta_o\right)$. Additionally, a number N_{out} of QPs can simulate multi-variable functions $f(\boldsymbol{s}) = (f_1(\boldsymbol{s}), \ldots, f_{N_{out}}(\boldsymbol{s}))$. To do this, note that output neuron $o \in \{1, \ldots, N_{out}\}$ can be chosen to experience couplings and detuning $\{J_{oi}, \Delta_o\}$ and single qubit rotations in Eq. (11) approximating $f_o\left(\sum_i J_{oi} s_i - \Delta_o\right)$. Notice that H_P creates entanglement between the input and the output. Indeed, the CNOT gate is a special case of H_P with additional single-qubit rotations since H_P can produce a control-Z gate (see Appx. 14 for details). Thus, a series of perceptrons can approximate any multi-variable function as long as the function's components have a Fourier series.

III. QUANTUM PERCEPTRONS ARE UNIVERSAL QUANTUM COMPUTERS

Sec. II C illustrates how one may use a QP with rotation on the output qubit to approximate any single variable function. We also discussed how a series of QPs can approximate multi-variable functions. In other words, a QP exhibits the same computational complexity as a classical perceptron. In this section, we show that if one equips the QP with rotations on both the input and the output, it can carry out universal quantum computation. That is, a QP's computational complexity exceeds that of its classical counterpart at the expense of single-qubit rotations.

To show that a QP can achieve universal quantum computation, it suffices to establish that the QP contains a universal set of gates. For concreteness, we show that a QP contains the Clifford+T universal gate set [20]. We need to show that (i) a QP can realize the identity gate; (ii) a QP can entangle any two qubits i, j through a control-Z gate where $CZ_{ij} = e^{i\pi|1\rangle\langle 1|_i \otimes |1\rangle\langle 1|_j}$, and that (iii) the QP can realize any single qubit rotations includ-

ing the Hadamard- (H), phase- (P), and T-gates. For simplicity, let $\tau = 1$.

Identity gate. We here show that any input qubit i Eq. (4) coupled with rotations can realize the trivial unitary $\mathbb{1}_i$, where $= |0\rangle\langle 0| + |1\rangle\langle 1|$ is the identity operator. To achieve this, we must eliminate the interactions between the input i and the output. We do this by introducing single-qubit rotations on qubit i.

To make this clear, let us consider a system that undergoes N_r rotations $\{R_n\}_{n=1}^{N_r}$. The leading-order average Hamiltonian H_{av} is a weighted average

$$H_{av} = \sum_{k=1}^{N_r} \tau_k \tilde{H}_k, \tag{12}$$

where τ_k is the duration between the $(k-1)^{th}$ and k^{th} rotations, which are denoted R_{k-1} and R_k. Here, $\tilde{H}_k = (R_{k-1}\dots R_1)^\dagger H_s (R_{k-1}\dots R_1)$ is the toggling frame Hamiltonian. We call H_s the native Hamiltonian of the system. This paradigm of the toggling frame and native Hamiltonian is called average Hamiltonian theory [21].

We now apply this framework to achieve an identity gate. If we want input i to decouple from the computation, we can rotate qubit i by a series of π-rotations along the x-axis. That is, we choose $R_k = e^{-i\pi/2\sigma_i^x}$, and $\tau = 1/N_r$. To see this, note that the native Hamiltonian between qubits i and o is $H_s = J_i\sigma_i^z\sigma_o^z$. But $R_k^\dagger \sigma_i^z R_k = -\sigma_i^z$ such that $R_k^\dagger H_s R_k = -H_s$. Rotating at each time increment τ produces the average Hamiltonian

$$H_{av} = \frac{1}{N_r}\left(\sum_{k \text{ even}} H_s + \sum_{k \text{ odd}} -H_s\right) = 0 \tag{13}$$

for even N_r. Thus, through rotation, we can realize identity gates on average. In Appx. B 1 we show that identity gates are readily realized exactly.

□

Two-qubit entangling gates. We now demonstrate that a QP can implement two-qubit entangling gates between any pair of qubits. Evidently, Eq. (4) tells us that any input and the output can interact via an entangling gate. Suppose for a moment that only input i interacts with the output. Introducing a rotation $R_z(\frac{\pi}{2}) = e^{-i\frac{\pi}{4}\sigma_i^z}$ and choosing $\Delta_o = -\pi/4$, $J_i = -\pi/2$ with $\Omega_o = 0$ results in the evolution

$$\begin{aligned}\exp\left\{-i\frac{\pi}{4}\left(\sigma_i^z + \sigma_o^z - \sigma_i^z\sigma_o^z\right)\right\} &= e^{-i\pi/4} e^{i\pi|1\rangle\langle 1|_i \otimes |1\rangle\langle 1|_o} \\ &= e^{-i\pi/4} CZ_{io}\end{aligned} \tag{14}$$

where we have used the observation $2|1\rangle\langle 1|_i = \mathbb{1}_i - \sigma_i^z$ [22]. Thus, up to an arbitrary phase, a QP plus rotations entangles the input and the output via a CZ gate.

The inputs, however, do not directly interact via Eq. (4). Since inputs i, j interact with the output, they can interact with each other via a second-order interaction. To see this, consider expanding the unitary

$$U_P = e^{-iH_P} = \mathbb{1} - iH_P - \frac{1}{2}H_P^2 + \dots. \tag{15}$$

Notice that the term H_P^2 contains terms of the form $J_iJ_j\sigma_i^z\sigma_j^z$, which generates entanglement between the inputs. However, H_P^2 creates all-to-all interactions in an uncontrolled fashion. We can also generate a more controllable two-qubit gate between two inputs. For instance, suppose that all but two inputs are rotated as in Eq. (13) and that the output experiences fields $\Omega_o \gg J_{1,2}$ with $\Delta_o = 0$. The QP then experiences the perturbative Hamiltonian

$$H = -\Omega_o\sigma_o^x + J_1\sigma_1^z\sigma_o^z + J_2\sigma_2^z\sigma_o^z = H_0 + V. \tag{16}$$

The low-energy eigenstate of H_0 is $|+\rangle_o$. If we initialize the QP with the output qubit in this state, then the input experiences the effective Hamiltonian (see Appx. B 2 for details)

$$H_{\text{eff}}^{\text{input}} = -\Omega_o + \sum_{ij} \frac{J_iJ_j}{\Omega_o}\sigma_i^z\sigma_j^z. \tag{17}$$

Thus, we have an Ising-like interaction on qubits 1 and 2 mediated by the output qubit o.

We can also utilize the above methodology to distill a control-Z operation among the inputs. Let $J_1 = \sqrt{\pi/4} = -J_2$. Suppose that the inputs evolve under Eq. (17) for a total time $1/\Omega_o$. Let input rotate along the z-axis by $R_z(-\pi/2)$. The total evolution of the inputs approximates CZ_{12} up to an arbitrary phase (see Appx. B 2 for details). □

Single-qubit gates. Finally, the construction of single-qubit gates follows from introducing single-qubit rotations on both the input and the output qubits. In fact, to complete the Clifford+T gates, it suffices to use rotations of the form $R_z(\pi/4)$ and $R_x(\pi/4)$ [23]. In Appx. B 3 we show that indeed $T = R_z(\pi/4)$, $P = T^2$, and $H = e^{-i\pi/2}R_x(\frac{\pi}{4})^2 R_z(\frac{\pi}{4})^2 R_x(\frac{\pi}{4})^2$. □

The three components above prove that QPs are universal for quantum computation. We now extend our study of QPs with a finite number of single-qubit rotations. We demonstrate the trainability and expressivity of QPs in the language of variational quantum algorithms. We highlight QPs' usefulness in quantum information processing applications.

IV. LEARNING WITH QUANTUM PERCEPTRONS

Thus far, we have been using QPs to realize known functions or quantum circuits. We now explore using the Hamiltonian Eq. (4) and single-qubit rotations in a machine-learning algorithm. In particular, we recast QPs

into a variational circuit. To do this, we can view the rotations as single-qubit unitaries and the evolution of the Hamiltonian as an entangling unitary. If we tune the rotation angles for each rotation, as well as the Hamiltonian parameters in Eq. (4) between the rotations, then we effectuate the unitary evolution

$$U(\boldsymbol{\theta}) = \prod_{l=1}^{L} U_l^2 e^{-i\tau H_P^l} U_l^1, \tag{18}$$

where L is half the number of rotations applied and is equivalent to the notion of circuit depth. Likewise, τ is a fixed time interval, and

$$H_P^l = \Omega_o^l \sigma_o^x + \left(-\Delta_o^l + \sum_{i=1}^{N} J_i^l \sigma_i^N \right) \sigma_o^z, \tag{19}$$

is the evolution Hamiltonian between two rotations with variational parameters $\{\Omega_o^l, \Delta_o^l, J_i^l\}$. The unitaries $U_l^{1,2}$ denote the single-qubit rotations. We choose an Euler representation of these rotations such that

$$U_k^l = \prod_{i=0}^{N+1} e^{-i\gamma_{ki}^l \sigma_i^z} e^{-i\beta_{ki}^l \sigma_i^x} e^{-i\alpha_{ki}^l \sigma_i^z} \tag{20}$$

with variational parameters $\{\gamma_{ki}^l, \beta_{ki}^l, \alpha_{ki}^l\}$ with $k = 1, 2$. We denote $\boldsymbol{\theta}$ as the vector of all variational parameters.

Let us now use a QP for learning. In the simple case that we seek a single output, we initialize the inputs and output qubits in the state $|0\rangle^{\otimes N+1}$ and utilize the QP to produce a network output by measuring an observable $\hat{\mathcal{O}}$ on the qubits after the evolution. The task is to minimize the mean-squared loss

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{2} \left(\langle 0|U^\dagger(\boldsymbol{\theta}) \hat{\mathcal{O}} U^\dagger(\boldsymbol{\theta})|0\rangle - \hat{\mathcal{O}}_0 \right)^2 = \frac{1}{2} \varepsilon(\boldsymbol{\theta})^2 \tag{21}$$

where $\hat{\mathcal{O}}_0$ is the target value and ε is the residual error. The parameters of the QP are then optimized using gradient descent

$$\boldsymbol{\theta}(t+1) = \boldsymbol{\theta}(t) - \eta \frac{\partial \mathcal{L}}{\partial \boldsymbol{\theta}(t)} \tag{22}$$

where η is the learning rate, and t denotes the tth training epoch.

Similarly, if instead of looking to obtain a single output value $\hat{\mathcal{O}}_0$, we are given a data set $\mathcal{D}$ with data points $\boldsymbol{x}$ and labels $\tilde{y}(\boldsymbol{x})$, the data can be loaded into the input qubits via a feature map $\boldsymbol{x} \to |\phi(\boldsymbol{x})\rangle|0\rangle$, where the last ket represents the output qubit. Then, the QP outputs

$$y_{\boldsymbol{\theta}}(\boldsymbol{x}) = \langle 0|\langle \phi(\boldsymbol{x})|U^\dagger(\boldsymbol{\theta}) \mathcal{O} U^\dagger(\boldsymbol{\theta})|\phi(\boldsymbol{x})\rangle|0\rangle \tag{23}$$

and the task becomes minimizing the loss

$$\mathcal{L}(\theta) = \frac{1}{2} \sum_{\boldsymbol{x} \in \mathcal{D}} \left(\tilde{y}(\boldsymbol{x}) - y_{\boldsymbol{\theta}}(\boldsymbol{x}) \right)^2 = \frac{1}{2} \sum_{\boldsymbol{x} \in \mathcal{D}} \varepsilon_{\boldsymbol{x}}(\boldsymbol{\theta})^2 \tag{24}$$

where $\varepsilon_{\boldsymbol{x}}$ is the residual error for data point $\boldsymbol{x}$.

A. The quantum neural tangent kernel of QPs

We now derive the quantum neural tangent kernel (QNTK) for the learning problem in 21. The QNTK, as detailed below, tells us about the rate of convergence of a QP towards a minimum of the loss function on the training set. Thus, the QNTK measures the trainability of our algorithm [24]. However, we highlight that the QNTK is purely an analytical tool hereby used only to compare QPs to other variational quantum algorithms (VQA) in QML. Whether the QNTK represents the training behavior in practice is yet to be determined.

The QNTK is defined as

$$K(\boldsymbol{\theta}) \equiv \sum_i \frac{\partial \varepsilon}{\partial \theta_i} \frac{\partial \varepsilon}{\partial \theta_i} \tag{25}$$

where i indexes the variational parameters. The QNTK is a positive real number that determines the evolution of the residual error ε since $\delta\varepsilon = -\eta K \varepsilon$ in the limit $\eta \to 0$. It is worth noting that, in the more general classification case Eq. (24), the QNTK is a positive-semidefinite matrix.

We will now analyze the so-called "lazy-training regime". In this regime, K becomes frozen to its average $\bar{K}$ over the QP's parameters when $\eta \to 0$. Following the techniques introduced in [25], we obtain

$$\bar{K} = \frac{6L(N+1)}{D^2 + D} \left(\frac{D\mathrm{Tr}(\hat{\mathcal{O}}^2) - \mathrm{Tr}^2(\hat{\mathcal{O}})}{D^2 - 1} \right) \tag{26}$$

where $D = 2^{N+1}$ is the size of the Hilbert space of the QP. For a full derivation of this result see Appx. C.

In this lazy-training regime, the residual error decays from an initial value $\varepsilon(0)$ to

$$\varepsilon(t) = \varepsilon(0) e^{-t\eta\bar{K}} \tag{27}$$

after t steps of gradient descent. Note that in the case that $L \approx D^2/(N\eta \mathrm{Tr}(\hat{\mathcal{O}}^2))$ the residual error in Eq. (27) decays with an exponential of order $\mathcal{O}(1)$. Since usually $\hat{\mathcal{O}}$ is a Pauli string such that $\mathrm{Tr}(\hat{\mathcal{O}}^2) \sim D$, as long as $L \sim D$ the kernel is of order $\mathcal{O}(1)$. That is, the training loss decays exponentially in $\mathcal{O}(\eta^{-1})$ iterations.

We numerically corroborate the results of Eq. (26) by training an $N = 3$ QP to produce a state such that

$$\langle X_i X_j \rangle + \langle Z_i Z_j \rangle = 2 \tag{28}$$

where $i, j \leq N$ are two inputs. Note that Eq. (28) is satisfied for a state with perfect correlations in both the x- and z-bases, such as the Bell-pair $(|00\rangle_{ij} + |11\rangle_{ij})/\sqrt{2}$. Thus, we interpret this task as searching for a state on two inputs that saturates the CHSH inequality [26].

Fig. 4 (A)-D shows the results of running the task in Eq. (26). For all experiments $\tau = 1$. The top panel shows the average kernel $\bar{K}$ obtained by calculating the residual error as $\epsilon = 2\sqrt{\mathcal{L}(\boldsymbol{\theta})}$, with $\mathcal{L}$ defined in 21, for training

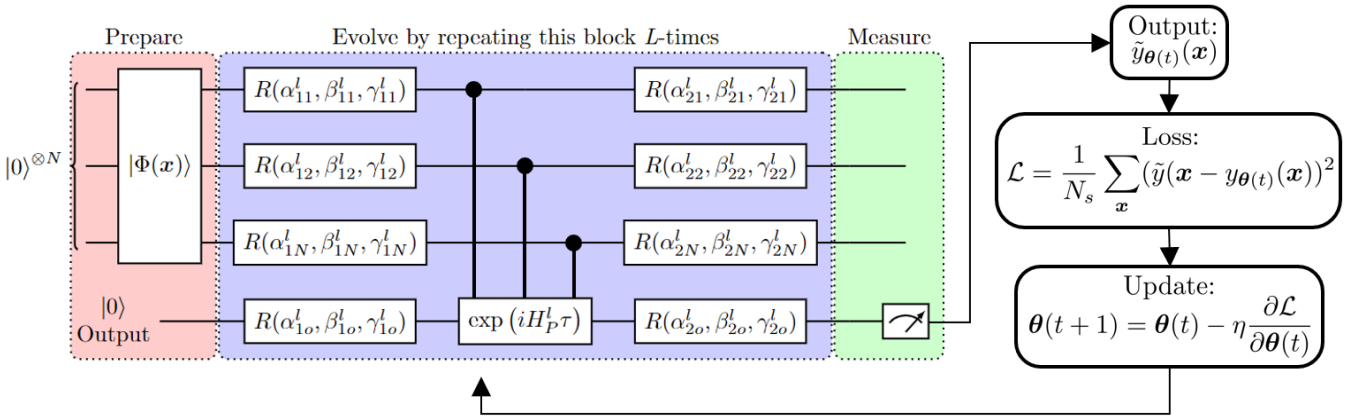

FIG. 3. Circuit representation of a quantum perceptron (QP). This circuit representation is just a visualization, but a QP uses the inherent dynamics of systems obeying H_P. Thus, a QP is closer to an analog computer than a digital one. On the first stage (left, red), we prepare an input $\boldsymbol{x}$ into the N qubit state $|\Phi(\boldsymbol{x})\rangle$, and the output qubit is in state $|0\rangle$. In the second stage (center, blue), the input-output system evolves via a series of single-qubit rotations along the x- and z-axes as in Eq. (20), and the unitary prescribed by the Hamiltonian H_P in Eq. (4) for a time τ. We chose this diagrammatic representation of $e^{iH^l_P\tau}$ to highlight that each input qubit contributes to the exact value of the z-field on the output qubit via the linear combination in Eq. (5). The variational parameters are $\boldsymbol{\theta} = \{\gamma^l_{ki}, \beta^l_{ki}, \alpha^l_{ki}, \Omega^l, \Delta^l, J^l_i\}$, After L blocks of evolution (right, green), the output qubit, and possibly some of the input qubits, are measured to form the output $y_{\boldsymbol{\theta}}(\boldsymbol{x})$ of the QP. The output is used to calculate a loss function comparing $y_{\boldsymbol{\theta}}(\boldsymbol{x})$ to the target function $\tilde{y}(\boldsymbol{x})$. We train the QPs parameters via gradient descent.

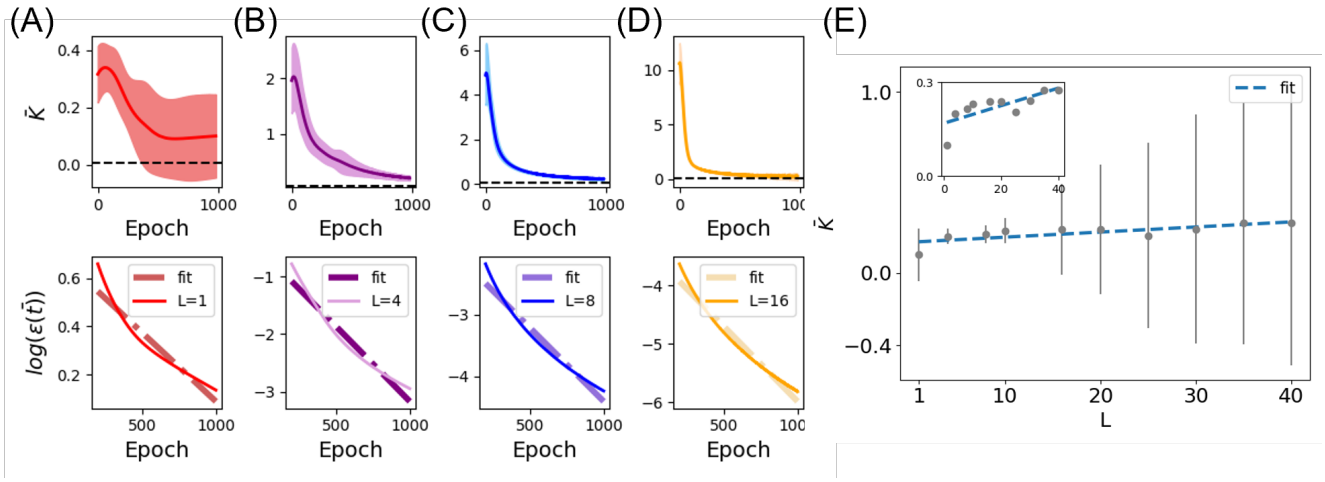

FIG. 4. (A)-(D) Average kernel and residual error of the task in Eq. (28) of finding states that break the CHSH-inequality with a QP of depths $L = 1, 4, 8$ and 16 and with $N = 3$ and $\tau = 1$. The top panel shows $\bar{K}$ obtained after 40 instances of the learning procedure. Here $\bar{K}$ is calculated using Eq. (27). The black line represents the prediction in Eq. (26) obtained from assuming the QP approximates a scrambling unitary. $\bar{K}$ resembles the prediction for increasing values of L. The bottom panel shows the average log of the residual errors. A linear fit shows that the log error more closely follows an exponential decay for $L \sim 2^{N+1}$. (E) Values of $\bar{K}$ as a function of L. The average is over 40 trained task instances. We see that $\bar{K}$ scales linearly with L, validating our findings in Eq. (26). The inset in (E) shows $\bar{K}$ without error bars. The error bars in (E) increase due to overfitting since we expect the task in Eq. (28) to need short depths. Indeed we see that at $L \sim 5$, the error bars decrease before increasing again.

epoch t and using Eq. (27) as a fit. $\bar{K}$ converges to the prediction in Eq. (26) with $\mathrm{Tr}(\hat{O}^2) = 2D$ for $L \sim D$, The variance on $\bar{K}$ is limited by the variance in the residual error. The bottom panel of 4 shows the log of the residual errors. A linear fit shows that the errors follow Eq. (27) for larger L.

Fig. 4 (E) corroborates that $\bar{K}$ is indeed linear with L. Error bars represent the standard deviation of $\bar{K}$ which in [25] is denoted $\Delta\bar{K}$. We note that $\Delta\bar{K}$ grows for increasing L. Growth of $\Delta\bar{K}$ is also reported in [25] where

$\Delta\bar{K} \sim \sqrt{L}$ and thus $\Delta\bar{K}/\bar{K} \sim L^{-1/2}$. However, in our case $\Delta\bar{K}/\bar{K}$ does not vanish with growing L. This discrepancy is because while our circuit can readily approximate a scrambling unitary at the level of a 2-design, at the explored depths, it fails to approximate the unitary 4-design used to derive the behavior $\Delta\bar{K}/\bar{K} \sim L^{-1/2}$ in [25]. We hope that future work can explore the depth or connectivity needed for a QP, or series of QPs, to approximate a unitary 4-design.

Most importantly for us, the QP's QNTK in Eq. (26) scales with L and D similar to the kernels of the more complicated VQAs in [24, 25]. Thus, as long as $L \sim D$, all these circuits have similar learning behaviors. Deep-circuits like this behave much like classical deep-neural networks [27]. However, while in the classical case, the kernel scales inversely with the number of neurons, deep VQAs' kernel scales inversely with an exponential of the number of neurons, thus resulting in slower training.

Lastly, we highlight that the QNTK formalism hinges on unrealistic assumptions that limit its scope in practical settings. Firstly, the QNTK assumes exact precision over the gradients of a VQA. Nonetheless, VQAs often experience training bottlenecks due to vanishingly small gradients [28, 29] whose estimation is not scalable [30]. Secondly, derivations of the QNTK often conclude that the algorithm is trainable whenever the depth scales exponentially with the system size. However, in this limit, it is impossible to ignore the effects of decoherence in currently available quantum devices and unaccounted for in the QNTK's derivation. Lastly, in deriving the QNTK, one assumes that the VQA's statistics are approximately equal to those of a scrambling unitary [31]. We give evidence to the validity of this assumption in Appx. C 1, where we numerically establish a connection between QPs and scrambling by looking at out-of-time-order correlators [32].

Nevertheless, it makes sense to use the QNTK as a purely an analytical tool which lets us compare QPs to other variational quantum algorithms (VQA) in QML.

V. APPLICATIONS OF QUANTUM PERCEPTRONS

We now apply QPs to common quantum computing and quantum machine tasks, specifically to measuring state inner products, entanglement detection, and quantum metrology.

We will use a QP without single-qubit rotations for the first task of state inner products. For the other applications, we use the fact that when Ω_o^l is the dominant parameter, the input qubits evolve under the effective Hamiltonian

$$H_{\text{eff}}^{\text{input}} = H^{\text{in}} \otimes |+\rangle\langle+|_o - H^{in} \otimes |-\rangle\langle-|_o \tag{29}$$
$$H^{\text{in}} = -\Omega_o^l + \sum_{ij} \frac{J_i^l J_j^l}{\Omega_o^l} \sigma_i^z \sigma_j^z,$$

as opposed to the more general H_P. Two salient features of Eq. (29) are worth noting. Firstly, Eq. (29) is valid when $\Omega_o^l \tau \ll 1$ and $J_i^l J_j^l / \Omega_o^l \ll 1$. Since the evolution is for short times, it is well suited for simulations using matrix product states (MPS) [33, 34]. In Appx. D, we show that Eq. (4), and hence Eq. (29), have a matrix product operator form H^{MPO} with bond dimension $d = 2$ such that the unitary generated by Eq. (29) can be approximated as

$$U \approx \mathbb{1} - i\tau H^{\text{MPO}} - \frac{\tau^2}{2} \left(H^{\text{MPO}}\right)^2, \tag{30}$$

which has bond dimension d^2. As a result, our simulations scale as $\mathcal{O}(Nd^{2L})$ which is more efficient than exact simulations for $L < N/2$. Since our simulations also require gradient-descent capabilities, we use TensorLyQuantum [35], a pure-PyTorch package for QML simulations.

Secondly, from Eq. (29) the Hamiltonian on the input (H^{in}) changes sign depending on whether the output is initialized in the state $|+\rangle_o$ or $|-\rangle_o$. Reversing the output state from $|+\rangle_o$ to $|-\rangle_o$ is equivalent to time-reversing the evolution of the input qubits. We will exploit this fact in Sec. V D.

A. Perceptrons for measuring state inner products

One important problem in QML is the measurement of kernel functions given a feature map. Suppose we have a data set $\mathcal{D}$ that is a subset of an ambient data set $\mathcal{X}$. Let ϕ be the feature map taking a data point $\boldsymbol{x}$ into a quantum state $|\phi(\boldsymbol{x})\rangle$. The kernel inherited by the featured map is defined as $\kappa(\boldsymbol{x}, \boldsymbol{y}) \equiv |\langle\phi(\boldsymbol{x})|\phi(\boldsymbol{y})\rangle|^2$ [36, 37]. The Representer Theorem [38] tells us that for a given machine learning task (a data set, a loss function, and a set of labels), the function g^* minimizing the loss can be written in terms of the kernel: $g^*(\boldsymbol{x}) = \sum_{\boldsymbol{y}} \alpha_{\boldsymbol{y}} \kappa(\boldsymbol{x}, \boldsymbol{y})$. Thus, a central goal of QML is to measure such kernels efficiently. Note that measuring these kernels amounts to measuring the inner product between two states. Inner products can be measured using the SWAP-test [39], assuming one can perform long-range SWAP gates, a missing capability in most current quantum devices.

Let us first show how one could measure the inner product between two one-qubit states using a QP. Our QP has two input qubits $i = 1, 2$ encoding the data $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ in $\mathcal{D}$ into the one-qubit states $|\phi(\boldsymbol{x}_1)\rangle$ and $|\phi^*(\boldsymbol{x}_2)\rangle$. Explicitly,

$$|\phi(\boldsymbol{x}_1)\rangle_1 = a_0(\boldsymbol{x}_1)\,|0\rangle_1 + a_1(\boldsymbol{x}_1)\,|1\rangle_1$$
$$|\phi^*(\boldsymbol{x}_2)\rangle_2 = a_0^*(\boldsymbol{x}_1)\,|0\rangle_2 + a_1^*(\boldsymbol{x}_2)\,|1\rangle_2\,.$$

Here $|\phi(\boldsymbol{x}_1)\rangle$ lives on the first input qubit, and $|\phi^*(\boldsymbol{x}_2)\rangle$ on the second input qubit. Suppose that these two input qubits are connected to an output qubit via a Hamiltonian with $J_0 = -J_1 = J$, $\Delta_o = 0$, and $\Omega_o = \pi$ with

$J \gg \Omega_o$. Then, the system evolves to

$$\sum_{ij} e^{-i(J((-1)^i-(-1)^j)\sigma_o^z+\Omega_o\sigma_o^x)} a_i(\boldsymbol{x}_1)a_j^*(\boldsymbol{x}_2)|ij0\rangle$$
$$\approx \sum_{i=j} a_i(\boldsymbol{x}_1)a_j^*(\boldsymbol{x}_2)|ij1\rangle$$
$$+ \sum_{i\neq j} e^{-i(\pm 2J\sigma_o^z)} a_i(\boldsymbol{x}_1)a_j^*(\boldsymbol{x}_2)|ij0\rangle \tag{31}$$

Thus, the probability of the output qubit being on state $|1\rangle$ is precisely $P(1) = |\sum_i a_i(\boldsymbol{x}_1)a_i^*(\boldsymbol{x}_2)|^2 = |\langle\psi(\boldsymbol{x}_1)|\psi(\boldsymbol{x}_2)\rangle|^2$, which is the kernel $\kappa(\boldsymbol{x}_1, \boldsymbol{x}_2)$. Note that this results from the condition $J \gg \Omega_o$ since only when $i = j$ the coherent drive $\Omega_o\sigma_o^x$ can significantly act on the output qubit.

One can generalize to larger input sizes straightforwardly. Suppose that state $|\psi(\boldsymbol{x}_1)\rangle$ now extends over n qubits $i = 1, ..., n$ and that state $|\psi^*(\boldsymbol{y}_2)\rangle$ extends over also n qubits $i = n+1, ..., 2n$. If we use $J_i = -J_{i+n} = J$ and $J \gg \Omega_o = \pi$ with $\Delta_o = 0$, then the probability of the output being on state $|1\rangle$ is again $P(1) = |\langle\psi(\boldsymbol{x}_2)|\psi(\boldsymbol{x}_1)\rangle|^2$. Our approach does not rely on the long-range gates necessary for the SWAP test. Rather, it relies on the natural dynamics of the QP. Note that H_P treats all the input qubits as neighbors to the output qubit. Thus, the QP employs near-range dynamics as long as all the inputs can be placed close enough to the output without interacting with each other. Thus, implementing QPs relies on high-connectivity or programmable platforms. In Sec. VI), we propose implementing QPs in several platforms, such as superconducting qubits [40], cold-atom arrays [41–47] and nitrogen-vacancy centers in diamond [21, 48].

The scheme presented above is a particular case of the Hadamard test [49]. The Hadamard test uses an ancilla to measure $\langle\psi| U |\psi\rangle$ where U is some unitary and ψ a state. Including pulses in the input of a QP realizes the Hadamard test since pulsing effectuates any unitary U. The output qubit can then be used as the ancilla to read the desired expectation value.

B. Using QPs for energy measurement

Let us consider the case in Eq. (29) with the output initialized in the state $|0\rangle_o$. The output qubit's expectation of σ_o^z is

$$\langle\sigma_o^z(\tau)\rangle \approx 2\tau \langle\Phi(\boldsymbol{x})| H^{\text{in}} |\Phi(\boldsymbol{x})\rangle \tag{32}$$

where $\Phi(\boldsymbol{x})$ is the prepared state of the input in the QP, and H^{in} is the effective Hamiltonian on the input generated by the QP's single-qubit rotations and evolution under Eq. (29). We note that H^{in} is a generalized Heisenberg model because of the single-qubit rotations H_P is an Ising-type interaction [21]. The result in Eq. (32) tells us that a QP can measure the energy of a state under a Hamiltonian H^{in}.

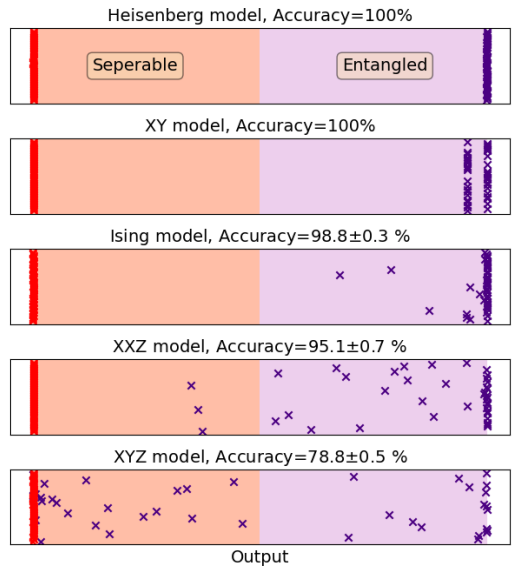

FIG. 5. We present the results of using QPs for entanglement detection. The purple crosses correspond to entangled states, while the red crosses correspond to separable states. The purple region is where the machine is trained to label entangled. The top panel shows the classification results on a test set made out of ground states of 1D Heisenberg models and their mean-field approximations. See the main text for a thorough discussion of the data set. For this task, $N = 4$ and $L = 4$ sufficed to give unit accuracy on the test set. The rest of the panels show the performance of the QPs on test sets consisting of ground states of other spin models of the form in Eq. (35) but are not included in the training set. The XY model corresponds to $J_z = h = 0$ and $J_x = J_y = J$, the Ising model to $J_z = J$, $J_x = J_y = 0$ and $h = 0.5$, the XXZ model for which $h = 0$ and $J_x = J_y$ and $J_z \neq 0$, and the XYZ model for which $h = 0$ and all couplings are independent. We see that the learned witness of the QPs can detect entanglement in different families of states. Moreover, the QP only labels separable states as separable. This correct labeling is due to the sufficiency of the witness.

We highlight that measuring energy as in Eq. (32) relies on measuring a single one-qubit observable as a function of time. This is far less expensive than traditional VQA [24, 25, 50] methods where $\mathcal{O}(N)$ two-point correlators need to be measured directly on a the many-body input system. Thus, our QP can measure the energy of a generalize Heisenberg Hamiltonians with enhanced precision and efficiency.

C. Using QPs for entanglement detection

We can use energy measurements under a many-body Hamiltonian for entanglement detection [51–53]. For certain Hamiltonians, the energy of its lowest-energy eigenvalues is lower than that of any separable state [53]. To explain this, let us define the energy gap of a generic

Hamiltonian $\hat{H}$ as

$$G = E_{\text{sep}} - E_0 \tag{33}$$

where E_0 is the ground state energy of $\hat{H}$, and E_{sep} is the lowest energy of a separable state under $\hat{H}$. The entanglement gap tells us about the properties of the Hamiltonian. For example, $G > 0$ if no state in the ground state manifold is separable. Every Hamiltonian with a positive entanglement gap defines an entanglement witness operator

$$W = \hat{H} - E_{\text{sep}}\mathbb{1} \tag{34}$$

since measuring $\langle\Psi|W|\Psi\rangle < 0$ on a state $|\Psi\rangle$ suffices to show that $|\Psi\rangle$ is entangled (i.e. not separable). We seek Hamiltonians with large entanglement gaps since these can detect entanglement even if the states have local imperfections or are at finite temperature [51]. Consider spin Hamiltonians of the form

$$H(J_{x,y,z}, h) = \sum_i \left(J_x\sigma_i^x\sigma_{i+1}^x + J_y\sigma_i^y\sigma_{i+1}^y + J_z\sigma_i^z\sigma_{i+1}^z\right) + h\sum_i \sigma_i^x \tag{35}$$

where $\sigma_i^y = i|1\rangle\langle 0| - i|0\rangle\langle 1|$. The Heisenberg model H_{Heis} is the case where $J_x = J_y = J_z = J$ and $h = 0$. In the case of the H_{Heis} the entanglement gap is $G \approx 0.77N$ so that if it is used for witnessing the quality of the witness improves linearly in N [52].

We now use a QP to produce an entanglement witness out of H^{in}. Note that the evolution in Eq. (29) and single-qubit rotations of a QP produce evolution under H^{in} on the input. Then, if Eq. (32) has a negative value for entangled states and a positive value for separable ones, H^{in} will have a nonzero entanglement gap and thus operate as a witness. Measuring the output of a QP then reveals the energy under this Hamiltonian using Eq. (32). This measurement indicates the value of a potential witness up to a constant factor.

In particular, we set $N = 4$ and proceed to generate a data set $\mathcal{D}$ consisting of two types of states. This first state type in $\mathcal{D}$ are ground states $|\text{GS}_J\rangle$ of 1D Heisenberg Hamiltonians with coupling-strength J sampled uniformly in the interval $[-1, 1]$. The second state type is the set of mean-field approximations $|\text{MF}_J\rangle$ of each ground state $|\text{GS}_J\rangle$, which can be written as the separable states $|\text{MF}_J\rangle = \bigotimes_{i=1}^N \left(\cos\theta_i\,|0\rangle + e^{i\phi_i}\sin\theta_i\,|1\rangle\right)$. The parameters $\{\theta_i, \phi_i\}$ are searched using gradient descent to minimize the energy $E_{\text{sep}} = \langle\text{MF}_J|H_{\text{Heis}}|\text{MF}_J\rangle$. Our data set consists of 200 ground states labeled with a value "-1" for entangled and 200 mean-field states labeled with a "1" as separable. Thus, $\mathcal{D} = \{(|\text{GS}_J\rangle, -1), (|\text{MF}_J\rangle, 1)\}_J$. Each state $|\psi\rangle$ in the data set is passed through a series of QPs with $\tau = 0.01, 0.02, \ldots, 0.1$ and fixed $\Omega_0^l = 50$. For each QP, we measure Eq. (32) and collect a vector $\boldsymbol{r}(\psi) = (\langle\sigma^z(0.01)\rangle, \ldots, \langle\sigma^z(0.1)\rangle, 1)$. The last entry of $\boldsymbol{r}(\psi)$ is a bias term. An output is produced by post-processing the measurements to obtain

$$\tilde{y}(\psi) = \text{softsign}(\boldsymbol{w}\cdot\boldsymbol{r}(\psi)) \tag{36}$$

where $\boldsymbol{w}$ is a vector of optimization parameters and $\text{softsign}(x) = x/(1 + |x|)$. We calculate the square-loss in Eq. (24) by summing over the states $\psi \in \mathcal{D}$. All the QPs have the same parameters, and only τ differs. The loss minimizes over the QP parameters and the vector $\boldsymbol{w}$. We choose to measure $\langle\sigma_o^z(\tau)\rangle$ for a variety of τ to give the post-processing more information to pin-point the energy of the input state under H^{in}.

Fig. 5 shows the results of using QPs for entanglement detection. The top panel shows the values of $\tilde{y}_{\boldsymbol{\theta}}$ for test states obtained from a Heisenberg model. We measure the accuracy as the percentage of test states that satisfy $|y(\psi) - \tilde{y}_{\boldsymbol{\theta}}(\psi)| < 1.0$. We note that $L = 4$ was sufficient to show unity accuracy on the Heisenberg test set.

The bottom five panels in Fig. 5 show the testing results against ground states (and their mean-field approximations) of other 1D spin models not included in the training set. In particular, we tested against the XY model for which $J_z = h = 0$ and $J_x = J_y = J$ [54], the Ising model for which $J_z = J$, $J_x = J_y = 0$ and $h = 0.5$ [55], the XXZ model for which $h = 0$ and $J_x = J_y$ and $J_z \neq 0$, and the XYZ model for which $h = 0$ and all couplings are independent [56]. All nonzero couplings $J_{x,y,z}$ were sampled uniformly on the interval $[-1, 1]$. The test sets consisted of 200 states each. Note that our QPs learned to classify all separable states for all models, while only some entangled states were classified correctly. This correct labeling is because entanglement witnessing is a sufficient and not necessary test of entanglement. We choose to train the QP using the ground states of Heisenberg Hamiltonians since we know that the Heisenberg model has a large energy gap suitable for witnessing. However, we could have also chosen to train using other ground states. Fig. 5 suggests that QPs trained on the Heisenberg model result in a powerful witness that can detect entanglement in other families of states not seeing during training.

The results in this section showcase the power of QPs for entanglement detection, a critical task in quantum computing and information. Thus, this section motivates using QPs to measure global dynamical quantities of quantum many-body systems.

D. Using QPs for quantum metrology

From Eq. (29), we see that flipping the output qubit from $|+\rangle_o$ to $|-\rangle_o$ changes the sign of the Hamiltonian H^{in} on the input qubits. As a result, a QP performs time-reversal operations on the input qubits. This time-reversal capability is an example of a well-known, ancilla-assisted operation similar to those in cold atoms in cavities [58–60]. With this observation and Sec. V C, we

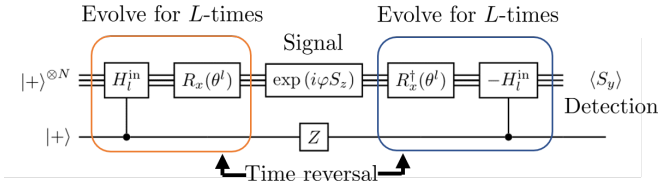

FIG. 6. We show the QP circuit representation for detecting a signal φ on the input qubits. The input is evolved from a coherent spin state (CSS) $|+\rangle^{\otimes N}$ to a spin-squeezed state via L applications of the Hamiltonian $H_{\text{in}} \propto S_z^2$. Here S_z is the total spin of the input qubits. We add global rotations to aid with squeezing. After the signal operates on the input, the output node changes from $|+\rangle_o$ to $|-\rangle_o$, effectively time-reversing the Hamiltonian evolution. The inverse of the global rotations is also applied. We measure $\langle S_y\rangle$ on the input. While for $\varphi = 0$ the state returns to the CSS with $\langle S_y\rangle = 0$, for small φ we obtain $\langle S_y\rangle = m\varphi N/2$ where m is the signal amplification.

can use the output qubit to steer and measure the input similar to in nitrogen-vacancy experiments [61].

Additionally, the effective Hamiltonian in Eq. (29) has the form (for $J_i^l = J^l$)

$$H_l^{\text{in}} = -\Omega^l + \frac{(J^l)^2}{\Omega_o^l}\sum_{ij}\sigma_i^z\sigma_j^z \propto S_z^2. \tag{37}$$

where $S_z = \frac{1}{2}\sum_{i=1}^N \sigma_i^z$ is the total input spin in the z-axis. This non-linear Hamiltonian is known to create highly entangled states by spin-squeezing initially separable states [58, 59, 62].

Let us now put together these two observations of our QPs dynamics. Highly entangled states are sensitive to small perturbations. Time-reversing entangling operations can help us measure the perturbation. The recently proposed signal amplification time-reversal interactions protocol (SATIN) illustrates this idea. In SATIN, a simple initial state is evolved to be highly entangled via a Hamiltonian of the form S_z^2 — called the Lipkin-Meshkov-Glick (LMG) [63, 64] — before interacting with a signal to be measured φ. Subsequently, the entangled state is time-reversed by evolving under $-S_z^2$. The final state is displaced from the original one, and under certain conditions, the signal φ is amplified so that it can be measured [62]. Concretely, as in [62], the perturbation is a global rotation along the y-axis $e^{-i\varphi S_y}$ where $S_y = \frac{1}{2}\sum_{i=1}^N \sigma_i^y$. The initial state is the coherent spin state (CSS) $|+\rangle^{\otimes N}$ for which $\langle S_y\rangle = 0$. After the time-reversal, however, if $\varphi \neq 0$, the state does not return to the CSS and instead [62]

$$\langle S_y\rangle = m\varphi N/2. \tag{38}$$

Here, $m \geq 1$ is called the amplification of the perturbation φ, and we denote $S_0 = N/2$.

In a QP, we achieve time reversal by switching the state of the output qubit from $|+\rangle_o$ to $|-\rangle_o$ or vice-versa. Since Ω_o^l is large, the evolution time τ must be chosen

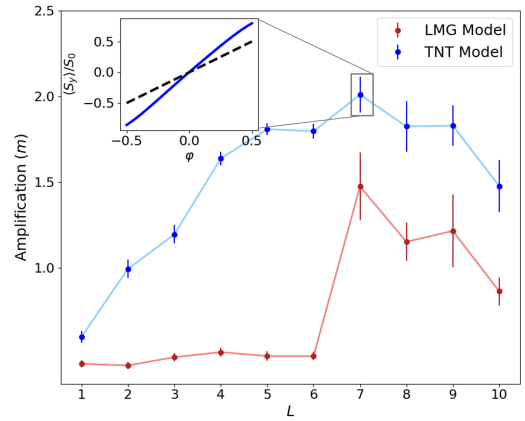

FIG. 7. We show the variational SATIN protocol for metrology on an $N = 10$ QP. We obtain the amplification m by measuring the input's total spin $\langle S_y\rangle = m\varphi S_0$ where $S_0 = N/2$ as in Fig. 6. We plot m as a function of the number of variational layers L. We present the results for the LMG model without global rotations along the x-axis (red lower curve) and the TNT model with global rotations along the x-axis (blue higher curve). As expected, the TNT model achieves a higher amplification m. The maximum m is 2.01, corresponding to a metrological gain of 3.03 dB. Note that the gain is smaller than the maximum for small and large values of L. For those values of L, our Hamiltonian produces a state under and over-squeezed, respectively. The inset shows a typical curve of the obtained $\langle S_y\rangle$ as a function of φ for the TNT model at $L = 7$. We see that the $\langle S_y\rangle$ closely fits a linear function with a slope $m > 1$. Note that $m < 1$ is worse than doing nothing at all. We include the $m = 1$ line as a reference (dotted black line). Our maximum metrological gain of 3.03 dB is about 2 dB below the best achieved gain for $N \sim 10$ using maximally-entangled states in Rydberg tweezer arrays [57]. However, QPs Hamiltonian evolution is less resource intensive as it relies on the dynamics of the system and not on adiabatic state preparation.

such that $\tau(J^l)^2/\Omega_o^l \ll 1$. Unlike the original SATIN proposal, a QP can perform intermittent single-qubit rotations. In this case, we set each rotation layer to perform global rotations along the x axis with angle θ^l. These rotations generate a so-called "twist-and-turn" (TNT) model where the state gets squeezed, then rotated, and so on [65]. We expect the TNT model to produce higher amplification than the LGM model [66]. We will allow the coupling and rotations per layer to be optimized. Thus, QPs realize a variational SATIN protocol.

While it is not novel to use time-reversal and spin-squeezing as tools for quantum metrology, we propose using a QP for metrology and casting measurement as a learning problem. This proposal is motivated by the QPs' ability to implement the TNT model and by the fact that one can use the output qubit to measure $\langle S_y\rangle$. To see this, One can imagine appending a layer to Fig. 6 composed of a $\pi/2$ rotation along the x-axis on the input, and a single use of H_P with $NJ\tau \ll 1$. In this case, the

probability of measuring the output qubit in the state $|+\rangle_o$ is (see Appx. E for details)

$$P_+ = (2J\tau)^2\langle S_y\rangle^2 = (2J\tau)^2(m\varphi N/2)^2. \tag{39}$$

Importantly, we expect to achieve higher precision and efficiency by using the output qubit for measurement than when measuring directly on the input qubits.

Before diving into the quantum metrology task results, we review the sensitivity to the signals measurable in different measurement protocols. For the typical case of using Ramsey interferometry [67] on N independent qubits to measure φ, the sensitivity of the probe is limited to $(\Delta\varphi)_{SQL} = 1/\sqrt{N}$. This result is known as the standard quantum limit (SQL) [68]. On the other hand, the precision can improve to $1/N$ by using a system of N highly entangled qubits, a result known as the Heisenberg Limit (HL) [69–71]. The HL, however, requires the production of a maximally entangled state, a feat only doable for a small number of qubits with significant experimental burdens [57, 72]. A more modest approach aims to achieve a Heisenberg scaling on the precision of b/N with $b > 1$ [73], which SATIN can accomplish.

The sensitivity of SATIN is approximated by $\delta\varphi = 1/(m\sqrt{N})$. As usual, we define the metrological gain of SATIN as

$$\mathcal{G} = \frac{(\Delta\varphi)^2_{SQL}}{\delta\varphi^2} = m^2. \tag{40}$$

Fig. 7 shows the results of the variational SATIN protocol for $N = 10$. We plot the amplification m as a function of L and the average results of 50 runs. Error bars represent one standard deviation from the mean. The LGM and TNT models achieve a maximum m at $L = 7$. This maximum is consistent with the results in [62] since the state before the perturbation is barely squeezed for small L (i.e., short evolution time). Likewise, for large L, the squeezing is so large that it washes out the signal. However, the highest amplification $m = 2.01$ is achieved with the TNT model leading to $\mathcal{G} = 3.01$ dB.

Interestingly, our protocol already outpaces SQL at $N = 10$, contrary to the predictions in [62]. We highlight that only Rydberg tweezer arrays have achieved a metrological gain $m > 1$ at $N \sim 10$ [57]. However, the QPs are less resource intensive that the approach in Ref. [57] which uses maximally entangled states. Thus, we expect that variational SATIN protocol on QPs improve on metrology protocols that use time-reversal operations. QPs' ability to steer the input qubits with a nonlinear Hamiltonian condition on the output qubit's state is helpful for metrology applications relevant to quantum information and processing.

VI. POTENTIAL EXPERIMENTAL IMPLEMENTATIONS OF QPS

Among the greatest motivations for pursuing this manuscript's manifestation of quantum perceptrons is that H_P is an interaction-rich Hamiltonian suitable for near-term devices. QPs consist of only two types of gates: analog evolution under H_P and single-qubit rotations. Thus, QPs can be implemented by several quantum computing and quantum simulation platforms in the near term.

In particular, as we shown in Sec. IV, we have represented the QP using a variational quantum circuit where the single rotation and the Hamiltonian can be realized via single rotation gate and entangling unitary respectively (see Fig. IV). This provides a straightforward protocol to implement the QP in gate based highly controllable and pulse-equipped quantum hardware such as superconducting qubit[40, 74].

However, the QP can be realized in other analog quantum hardware as well. For instance, QPs could be realized with Rydberg atom arrays that contain two atomic species [41], with one species being used for measurement and manipulation of the other. Currently, however, this platform can only perform global rotations, which may decrease the expressivity of the implementation but can still realize tasks such as the one in Sec. V D. A variation on this platform with two Rydberg states and one atomic species can also be used to implement QPs. For example, experiments in [42, 43] ensembles of Rubidium-87 Rydberg atoms are used where two Rydberg excitations ($60S$ and $60P$) are used so that atoms in the $60S$ state interact strongly with atoms in the $60P$ state while inter-state interactions are negligible. For example, a QP could be realized by encoding all inputs in the $60S$ state and retrieving the output in the $60P$ state.

Recent developments in cold atom arrays' reconfigurable geometries [44] and movable trapped ions [45, 46] can also implement QPs. In both platforms, atoms can be moved around to create effective interactions. As a result, the input qubit of a QP can be made to interact with the output qubits one at a time, avoiding input cross-talk. The separation between the input and output qubits also allows for flexible implementation of single-qubit control.

Atomic arrays with wavelength spacing, such as those generated by optical lattices, are also promising platforms for QPs. The strong, short-range dipole-dipole interactions of these structures endow them with rich, tuneable atom-atom interactions [75]. In the case of multiple atomic species or states, these systems' interaction length and coherence times can be expanded through array-mediated atom-atom dynamics [76], and analogous implementations have also been proposed for atomic dimers [77]. Two-dimensional lattices of wavelength spacing have particular potential due to their ability to interact deterministically with a single photon [78] and their recent experimental implementations [47].

Passing from atomic quantum computers to solid-state platforms, we note color centers in diamonds as a promising system for implementing QPs. For instance, H_P resembles the so-called "star model" used to describe how an electronic spin in a nitrogen-vacancy center in

a diamond interacts with neighboring nuclear spins [48]. Thanks to recent developments in single-qubit rotation sequences [21, 79, 80], single-qubit control is attainable within the near feature. However, the downside of this platform is the lack of tunability of the coupling constants J_i in Eq. (4).

VII. CONCLUSION

In this manuscript, we introduce quantum perceptrons (QPs) based on the unitary dynamics of spin systems. We show that QPs are helpful for common quantum machine learning problems, such as measuring the inner product between two states. We prove that, by adding single-qubit control to our QPs, we can approximate classical perceptrons and thus approximate multi-variable functions using multiple QPs. The approximation is noise insensitive to changes in the input. Therefore, architectures built out of QPs promise to be noise-resilient. Employing this added control, we also prove that QPs are universal quantum computational units, establishing that a single QP offers far more computational complexity than its classical counterpart. We also discuss using QPs for machine learning tasks by optimizing the couplings and local rotations of a QP using gradient descent and how to represent QPs as a variational quantum circuit. We analytically compute the quantum neural tangent kernel (QNTK) of QPs. The QNTK characterizes the epochs needed to learn a particular task and thus is a metric of the trainability of the circuit. The QP's QNTK for deep-circuits is like the QNTK behavior of other more complicated variational quantum algorithms.

The studies in this manuscript address a question that has recently been debated as a fundamental one for QML: "what are good building blocks for quantum [machine learning] models?" [15]. Out QPs answer this question with a simple and modular architecture. This architecture is inspired by quantum simulation tools which may enable QPs to be scalable soon. Our QPs are also insensitive to noise in the input states (Sec. II C), and their "non-classicality" clearly stems from single-qubit pulses, as seen in Sec. III. Moreover, QPs are especially well-suited for macroscopic observable measurements of the input qubits using the output qubit as a probe. In particular, we showed that QPs could measure energies of many spin models (Sec. V B) and total spin (Sec. V D). Thus, QPs measure quantities accurately and efficiently that would otherwise be hard to compute directly on a many-body system. We argue that QPs are an essential step towards scalable quantum machine learning applications on near-term devices for the reasons highlighted above.

Nonetheless, a further work is needed to discover the full potential of QPs. Directly from this work, investigations into the trade-off between circuit depth and higher connectivity of qubits are of great interest. These studies can be done by either stacking QPs or connecting the inputs at specific intermediate steps. Our QNTK discussion leaves several open questions, such as whether the scrambling constraint can be removed in deriving the QNTK or how noise in the inputs or output affects the kernel. Large-scale QP implementations may also benefit from gradient-free training schemes such as those recently developed for other analog algoritms [81]. Another direction of interest is using QPs for distributed computing, where a quantum computer prepares a particular state, and a QP is used to encode macroscopic quantities of the state, which can then be transported to do further computation [82], or using the restricted connectivity of QPs as blocks in larger computation [83]. QPs may also be incorporated as subroutines of other variational quantum circuits along the lines of the measurement-based perceptrons in [84]. Recent developments in dynamically reconfigurable computing geometries may realize this process [44–46]. While in Sec. V our algorithms used short-time dynamics, the type of problems that long-term dynamics can solve remains an open question. These studies will likely benefit from an analytical understanding of statistical ensembles of many-body systems.

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Appendix A: Rotation sequences for activation function emulation

First, let us derive Eq. (6). Note that for any one of the basis states $|\boldsymbol{s}\rangle\,|0\rangle_o = |s_1, ..., s_N\rangle\,|0\rangle_o$, we have that Eq. (4) gives us the evolution

$$\begin{aligned} e^{-i\left(\Omega_o\sigma_o^x - \Delta_o\sigma_o^z + \sum_i J_i\sigma_i^z\sigma_o^z\right)}\,|\boldsymbol{s}\rangle\,|0\rangle_o &= |\boldsymbol{s}\rangle\, e^{-i\left(\Omega_o\sigma_o^x - \Delta_o\sigma_o^z + \sum_i J_i s_i\sigma_o^z\right)}\,|0\rangle_o \\ &= |\boldsymbol{s}\rangle\, e^{-i(\Omega_o\sigma_o^x + h(\boldsymbol{s})\sigma_o^z)}\,|0\rangle_o \end{aligned}$$

where $h(\boldsymbol{s}) = \sum_i J_i s_i - \Delta_o$. The evolution of the output is that of a rotation $e^{-i\omega\hat{\boldsymbol{n}}\cdot\boldsymbol{\sigma}}$ with $\hat{\boldsymbol{n}} = (\Omega_o, 0, h(\boldsymbol{s}))/\omega$, $\omega = \sqrt{\Omega_o^2 + h(\boldsymbol{s})^2}$, and $\boldsymbol{\sigma} = (\sigma_o^x, \sigma_o^y, \sigma_o^z)$. This unitary can be represented as

$$\mathbb{1}\cos(\omega) - i\hat{\boldsymbol{n}}\cdot\boldsymbol{\sigma}\sin\omega.$$

Multiplying this last equation with the vector $|0\rangle_o$ gives us

$$|\boldsymbol{s}\rangle \left(\left[\cos\omega(\boldsymbol{s}) - i\frac{h(\boldsymbol{s})}{\omega(\boldsymbol{s})}\sin\omega(\boldsymbol{s}) \right] |0\rangle_o + i\frac{\Omega_o}{\omega(\boldsymbol{s})}\sin\omega(\boldsymbol{s}) \right) |1\rangle_o .$$

If instead of a single ket $|\boldsymbol{s}\rangle$, we were given that the perceptron starts in the state $\sum_{\boldsymbol{s}} \psi(\boldsymbol{s})|\boldsymbol{s}\rangle|0\rangle_o$, since the unitary evolution is linear, we would obtain Eq. (6). Note that according to Eq. (6)

$$f(h(\boldsymbol{s})) = \arcsin \frac{1}{\sqrt{1+h(\boldsymbol{s})^2/\Omega_o^2}} \sin\left(\Omega_o\sqrt{1+h(\boldsymbol{s})^2/\Omega_o^2}\right) \tag{A1}$$

which is an oscillatory function on $h(\boldsymbol{s})$ and Ω_o and thus nonlinear on the input. See Fig. 1 in the main text for the shape of this function.

Now, we derive the activation function obtained from rotating the output qubit. Consider an activation function with a Fourier series valid on an interval $[-\pi, \pi]$

$$\begin{aligned} f(h) &= a_0 + \sum_{k=1}^{\infty} a_k \cos(kh) + \sum_{k=1}^{\infty} b_k \sin(kh), \\ a_k &= \frac{1}{\pi}\int_{-\pi}^{\pi} dh f(h)\cos(kh), \\ b_k &= \frac{1}{\pi}\int_{-\pi}^{\pi} dh f(h)\sin(kh). \end{aligned}$$

Accordingly, we can rewrite $f(h)\sigma_N^x$ as

$$\begin{aligned} f(h) &= a_0\sigma_N^x + \sum_k (a_k\cos kh + b_k \sin kh)\sigma_N^x \\ f(h) &= a_0\sigma_N^x + \sum_k \frac{a_k}{2}(\cos kh\sigma_N^x + \sin kh\sigma_N^y + \cos kh\sigma_N^x - \sin kh\sigma_N^y) \\ &\quad + \sum_k \frac{b_k}{2}(\sin kh\sigma_N^x + \cos kh\sigma_N^y + \sin kh\sigma_N^x - \cos kh\sigma_N^y) \\ &= a_0\sigma_N^x + \sum_k \frac{a_k}{2}(e^{-ikh/2\sigma_N^z}\sigma_N^x e^{ikh/2} + e^{ikh/2\sigma_N^z}\sigma_N^x e^{-ikh/2}) \\ &\quad - \sum_k \frac{b_k}{2}(e^{-i(kh/2+\pi/4)\sigma_N^z}\sigma_N^x e^{i(kh/2+\pi/4)} + e^{i(kh/2+\pi/4)\sigma_N^z}\sigma_N^x e^{-i(kh/2+\pi/4)}) \end{aligned}$$

Thus, one can write

$$\begin{aligned} U_f = e^{-ia_0\sigma_o^x}\Pi_k \exp\left\{-i\frac{a_k}{2}\left(e^{-ikh/2\sigma_o^z}\sigma_o^x e^{ikh/2\sigma_o^z} + e^{ikh/2\sigma_o^z}\sigma_o^x e^{-ikh/2\sigma_o^z}\right)\right\} \\ \Pi_k \exp\left\{i\frac{b_k}{2}\left(e^{-i(kh/2+\pi/4)\sigma_o^z}\sigma_o^x e^{i(kh/2+\pi/4)\sigma_o^z} + e^{i(kh/2+\pi/4)\sigma_o^z}\sigma_o^x e^{-i(kh/2+\pi/4)\sigma_o^z}\right)\right\} \end{aligned}$$

Let us denote $R_z(x) = e^{-ix/2\sigma_o^z}$, $H_o = \sigma_o^x$, and let $W_k(h), V_k(h)$ denote the exponentials above for a_k and b_k respectively;

$$W_k(h) = \exp\left\{-i\frac{a_k}{2}\left(e^{-ikh/2\sigma_o^z}\sigma_o^x e^{ikh/2\sigma_o^z} + e^{ikh/2\sigma_o^z}\sigma_o^x e^{-ikh/2\sigma_o^z}\right)\right\} \tag{A2}$$

$$V_k(h) = \exp\left\{i\frac{b_k}{2}\left(e^{-i(kh/2+\pi/4)\sigma_o^z}\sigma_o^x e^{i(kh/2+\pi/4)\sigma_o^z} + e^{i(kh/2+\pi/4)\sigma_o^z}\sigma_o^x e^{-i(kh/2+\pi/4)\sigma_o^z}\right)\right\} \tag{A3}$$

$$U_f = e^{-ia_0\sigma_o^x}\Pi_k U_k(h)V_k(h). \tag{A4}$$

Note that the Hamiltonians above are of the form $H = e^{ih\sigma_o^z}\sigma_o^x e^{-ih\sigma_o^z}$. If we had white noise on $h \to h + \chi$ where $\chi \sim N(0, s_z)$, the average Hamiltonian is (to second order in s_z)

$$\bar{H} \approx (1 - s_z^2)e^{ih\sigma_o^z}\sigma_o^x e^{-ih\sigma_o^z}. \tag{A5}$$

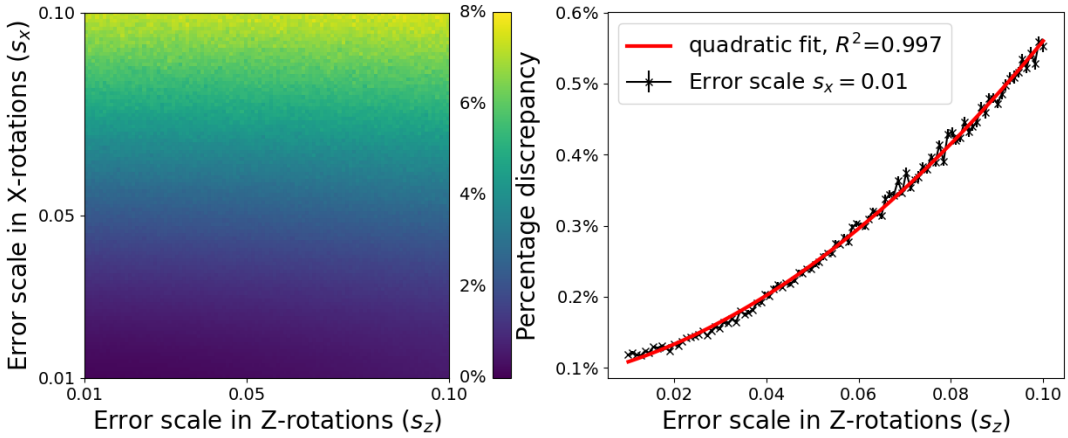

FIG. 8. Percentage discrepancy of approximating the sigmoid function with 15 Fourier terms. The left panel shows the discrepancy as a function of error scale in each type of rotation. The discrepancy is largely independent to the error scale s_z. The right panel shows the percentage discrepancy for fixed $s_x = 0.01$ and as a function of s_z. The error follows a quadratic prediction.

Thus, the Hamiltonian is robust to this error.

Let us now consider introducing zero-mean Gaussian noise to the Fourier coefficients a_k, b_k sampled from $N(0, s_x)$. We call s_x and s_z the error scales. Note that these labels highlight that the Fourier coefficients are related to evolving under σ_o^x (x-rotations), while h is related to evolving under σ_o^z (z-rotations). Fig. 8 shows the percentage discrepancy of calculating the sigmoid function as a function of the error scales. The percentage discrepancy is calculated as the average of the quantity

$$100 \int_{-3}^{3} dh \frac{|P_0(h) - \text{sigmoid}(h)|^2}{\text{sigmoid}(h)}. \tag{A6}$$

The average is taken over 50 instances of the errors on the coefficients and the field h. Note that here we chose the interval $h \in [-3, 3]$ and not $h \in [-\pi, \pi]$. This choice is based on the observation that our approximation will largely fail near the endpoints of the interval $[-\pi, \pi]$. This discrepancy is due to the fact that any finite Fourier series fails most largely at the endpoints (which in our case is $[-\pi, \pi]$).

In Fig. 8 the left panel shows that the discrepancy is largely independent to the scale s_z as expected. The right panel shows that the discrepancy follows a quadratic fit as a function of s_z and for fixed s_x.

Appendix B: Quantum perceptrons and universality

1. Supplementary calculation for identity gates

In Sec. III we showed that the rotation sequence

$$R_n = e^{-i\pi/2\sigma_k^x} \tag{B1}$$

for $n = 1, ..., N_r$ realizes, in average, the identity gate on qubit k. That is, the first moment of the effective Hamiltonian experience by qubit k vanishes. We can also show that this rotation sequence produces an effective Hamiltonian with vanishing higher moments. To see this, we rely on average Hamiltonian theory (for a comprehensive review of AHT see Appendix A in [21]). According to AHT, the evolution of a time-dependent Hamiltonian $H(t)$ generates can be also generated by a time-independent Hamiltonian $U = e^{-iH_{\text{eff}}}$ where $H_{\text{eff}} \approx \sum_{k=1}^{l} H_l$ where l is the truncation order and H_k is the kth moment contribution. Given a rotation sequence, the average Hamiltonian, or fist moment, is

$$H_{avg} = H_1 = \int_0^1 H(t)dt \tag{B2}$$

where $H(t)$ is the Hamiltonian at time t generated by the rotation sequence. In the case of Eq. (B1), as shown in Eq. (13), this average Hamiltonian vanishes. The next moment of this Hamiltonian is

$$H_2 = -\frac{i}{2}\int_0^1 dt_2 \int_0^1 dt_1 [H(t_2), H(t_1)] \tag{B3}$$

For the case of Eq. (B1) the commutators contain terms of the form $\pm[\sigma_i^z\sigma_o^z, \sigma_j^z\sigma_o^z]$ all of which vanish. Thus, $H_2 = 0$. Similarly, higher order moments contained nested commutators of Pauli-Z operators, all of which vanish.

2. Supplementary calculations for two-qubit gates

Let us now show that indeed the QP contains the control-Z gate between the input and output qubits. That is, we derive Eq. (14). Consider the case of only qubit i interacting with the output. For $J_i = -\frac{\pi}{4}$, $\Delta_o = -\pi/4$ and a rotation $R_z(\pi/2)$ on i we obtain

$$\begin{aligned} R_z^i\left(\frac{\pi}{2}\right) e^{-i\frac{\pi}{4}(\sigma_o^z - \sigma_i^z\sigma_o^z)} &= e^{-i\frac{\pi}{4}\sigma_i^z} e^{i\frac{\pi}{4}(\sigma_o^z - \sigma_i^z\sigma_o^z)} \\ &= e^{i\frac{\pi}{4}(\sigma_i^z + \sigma_o^z - \sigma_i^z\sigma_o^z)} \\ &= e^{i\frac{\pi}{4}(\mathbb{1}_i - \sigma_i^z)\otimes(\mathbb{1}_o - \sigma_o^z))} e^{-i\frac{\pi}{4}} \\ &= e^{-i\frac{\pi}{4}} e^{i\pi|1\rangle\langle 1|_i \otimes |1\rangle\langle 1|_o} \\ &= e^{-i\frac{\pi}{4}} CZ_{io} \end{aligned}$$

where we have used the fact that $2\,|1\rangle\langle 1| = \mathbb{1} - \sigma^z$.

We now turn to the derivation of two-qubit gates. Suppose we have $\Delta_o = 0$ and

$$H_p = -\Omega_o\sigma_o^x + \sum_j J_j\sigma_j^z\sigma_o^z = H_0 + V. \tag{B4}$$

where $\Omega_o \gg J_j$ so that V is a perturbation. Let's construct the effective Hamiltonian on the input layer. We use the Schieffer-Wolf transformation [85]. That is, we seek a unitary transformation S so that $H' = e^S H e^{-S}$ is diagonal to lowest order in V. We expect that since V is small, in the sense that $|\lambda/\Omega_o| \ll 1$ for any eigenvalue λ of V, S is also small. Using the Baker-Campbell-Hausdorff formula

$$H' = H_0 + V + [S, H_0] + [S, V] + \frac{1}{2}[S, [S, H_0]] + O(V^3). \tag{B5}$$

If we choose $[S, H_0] = -V$, then

$$H' = H_0 + \frac{1}{2}[S, V] + O(V^3) \tag{B6}$$

and thus to second-order perturbation theory, $[S, V]$ gives us the effective Hamiltonian. Suppose the eigenvectors and eigenvalues of H_0 are known so $H_0 = \sum_m E_m|m\rangle\langle m|$. Then, $[S, H_0] = -V$ gives us

$$\begin{aligned} \sum_m (SE_m|m\rangle\langle m| - E_m|m\rangle\langle m|S) &= -V \\ \sum_m E_m\left(\langle i|S|m\rangle\langle m|j\rangle - \langle i|m\rangle\langle m|S|j\rangle\right) &= -\langle i|V|j\rangle \\ \langle i|S|j\rangle &= \frac{\langle i|V|j\rangle}{E_i - E_j}. \end{aligned} \tag{B7}$$

This immediately gives us

$$\frac{1}{2}[S, V]_{ij} = \frac{1}{2}\sum_k \left(\frac{\langle i|V|k\rangle\langle k|V|j\rangle}{E_i - E_k} - \frac{\langle i|V|k\rangle\langle k|V|j\rangle}{E_k - E_j}\right) \tag{B8}$$

which is the result of second-order perturbation theory:

$$H' = H_0 + \frac{1}{2}\sum_{ij}\sum_k \left[\frac{\langle i|V|k\rangle\langle k|V|j\rangle}{E_i - E_k} - \frac{\langle i|V|k\rangle\langle k|V|j\rangle}{E_k - E_j}\right] |i\rangle\langle j| \tag{B9}$$

It is usual to ask that V is completely off-diagonal in the eigenbasis of H_0. Finally, let's go back to the specific H_0 and V in Eq. (B4). Notice that the eigenvectors of H_0 give us the basis $\{|+\rangle_o|\sigma\rangle, |-\rangle_o|\sigma\rangle\}_\sigma$ where $\{|\sigma\rangle\}$ is any eigenbasis on the input qubits. The energies are $-\Omega_o, \Omega_o$. To see that the Hamiltonian in eq. Eq. (B9) is indeed diagonal, consider the case that $|i\rangle = |+_o, \sigma\rangle$ and $|j\rangle = |-_o, \sigma'\rangle$. The terms in the brackets in Eq. (B9), for $|k\rangle = |k_1, \sigma''\rangle$ with $k_1 = \pm$, are easily seen to be zero since $\langle -_o|\sigma_o^z|-_o\rangle = \langle +_o|\sigma_o^z|+_o\rangle = 0$. Thus, we end up with

$$\begin{aligned} H' &= H_0 - |+\rangle\langle +|_o \otimes \sum_{ij} \frac{J_i J_j}{\Omega_o} \sigma_i^z \sigma_j^z + |-\rangle\langle -|_o \otimes \sum_{ij} \frac{J_i J_j}{\Omega_o} \sigma_i^z \sigma_j^z \\ &= |+\rangle\langle +|_o \otimes \left(-\Omega_o + \sum_{ij} \frac{J_i J_j}{\Omega_o} \sigma_i^z \sigma_j^z \right) + |-\rangle\langle -|_o \otimes \left(\Omega_o - \sum_{ij} \frac{J_i J_j}{\Omega_o} \sigma_i^z \sigma_j^z \right) \\ &= |+\rangle\langle +|_o \otimes H_{\text{eff}}^{\text{input}} - |-\rangle\langle -|_o \otimes H_{\text{eff}}^{\text{input}}. \end{aligned} \tag{B10}$$

This last equation leads to the result in Eq. (17).

Lastly, consider the case where only qubits $i = 1, 2$ participate in this effective Hamiltonian and that as this Hamiltonian acts, we also rotate the inputs by an $R_z\left(\frac{-\pi}{2\Omega_o}\right)$. Choosing $J_1 = -J_2 = \sqrt{\pi/4}$ we obtain the unitary

$$\begin{aligned} U_{12} &= R_z^1\left(\frac{-\pi}{2\Omega_o}\right) R_z^2\left(\frac{-\pi}{2\Omega_o}\right) e^{-i\Omega_o} e^{-i\frac{\pi}{4\Omega_o}\sigma_1^z\sigma_2^z} \\ &= e^{-i\Omega_o} e^{-i\frac{\pi}{4\Omega_o}(\sigma_1^z + \sigma_2^z - \sigma_1^z\sigma_2^z)} \\ &= e^{i\Omega_o} e^{-i\frac{-\pi}{4} CZ_{io}^{1/\Omega_o}}. \end{aligned}$$

If we perform this operation Ω_o times, then our gate is approximately CZ_{io} up to the phase $\pi\Omega_o/2 - \Omega_o^2$. Since H_{eff} is correct up to order $\mathcal{O}(\Omega_o^{-2})$, the total error is of order $\mathcal{O}(\Omega_o^{-1})$.

3. Supplementary calculations for single-qubit gates

The Clifford gates are the single-qubit Hadamard- (H), phase- (P) gates, and the two-qubit control-Z (CZ) gate. We augment the set $\{H, P, CZ\}$ with the so-called T gate to form a universal gate set [20]. The single-qubit gates are defined as follows

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad P = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}. \tag{B11}$$

We use the convention $R_{x,z}(\theta) = e^{-i\theta/2\sigma^{x,z}}$. Evidently $P = T^2$. Note that $T = e^{-i\pi/8} R_z(\pi/4)$. Lastly, a straightforward calculation shows that $H = -iR_x(\pi/2)R_z(\pi/2)R_x(\pi/2)$. Since $R_{x,z}(2\theta) = R_{x,y}(\theta)^2$, rotations $R_{x,z}(\pi/4)$ are all we need to produce the Clifford+T set. As long as those rotations are present in a QP, the QP can realize universal quantum computation.

Appendix C: Derivation of QP's quantum neural tangent kernel

Let us compute the derivatives of the residual error Eq. (21) with respect to the QPs parameters. To make things more convenient, we will use single-qubit rotations $e^{i\theta_i^l X_i}$ where X_i^l is a normalized, linear combination of Pauli operators on qubit i. We will also let $W_l = e^{-i\tau H_P^l}$.

If θ_i corresponds to one of the parameters in U_l^1, we denote it by $\theta_{l,i}^1$. The index l denotes the layer, and i the qubit. Then

$$\begin{aligned} \frac{\partial \varepsilon}{\partial \theta_{l,i}^1} &= \langle 0| U_1^{1\dagger} W_1^\dagger U_2^{2\dagger} \dots U_{l-1}^{1\dagger} W_{l-1}^\dagger U_{l-1}^{2\dagger} (-iX_i^l) W_l U_l^{2\dagger} \dots \hat{\mathcal{O}} \dots |0\rangle + \langle 0| \dots \hat{\mathcal{O}} \dots U_l^2 W_l U_l^1 (iX_l^i) \dots |0\rangle \\ &= -i \langle 0| U_{-,l,1}^\dagger \left[X_l^i, U_{+,l,1}^\dagger \hat{\mathcal{O}} U_{+,l,1} \right] U_{-,l,1} |0\rangle \end{aligned} \tag{C1}$$

$$U_{+,l,1}^\dagger = U_L^2 W_L U_L^1 \dots U_{l+1}^2 W_{l+1} U_{l+1}^1 U_l^2 \tag{C2}$$

$$U_{-,l,1}^\dagger = U_{l-1}^2 W_{l-1} U_{l-1}^1 \dots U_1^2 W_1 U_1^1. \tag{C3}$$

Similarly, if θ_i corresponds to one of the parameters in U_l^2, we get

$$\frac{\partial \varepsilon}{\partial \theta_{l,i}^2} = -i \langle 0| U_{-,l,2}^\dagger \left[U_{+,l,2}^\dagger \hat{\mathcal{O}} U_{+,l,2} \right] U_{-,l,2} |0\rangle \tag{C4}$$

$$U_{-,l,2} = U_l^2 W_l U_l^1 \dots U_1^1 \tag{C5}$$

$$U_{+,l,2} = U_L^2 W_L U_L^1 \dots U_{l+1}^2 W_{l+1} U_{l+1}^1 U_l^2. \tag{C6}$$

Lastly, if θ_i stands for a parameter of the Hamiltonian, we denote it as $\theta_{l,i}$ with $i = 1, \dots, N$ corresponding to a coupling and $i = N+1, N+2$ corresponding to Δ_l, Ω_l. We obtain

$$\frac{\partial \varepsilon}{\partial \theta_{l,i}} = -i \langle 0| U_{-,l}^\dagger \left[A_i, U_{+l}^\dagger \hat{\mathcal{O}} U_{+,l} \right] U_{-,l} |0\rangle \tag{C7}$$

$$U_{-,l} = W_l U_l^1 \dots U_1^2 W_1 U_1^1 \tag{C8}$$

$$U_{+,l} = U_L^2 W_L U_L^1 \dots U_l^2, \tag{C9}$$

where $A_i \in \{\sigma_i^z \sigma_o^z, \sigma_o^z, \sigma_o^x\}$.

The average QNTK, $\bar{K}$, splits into three parts. We call $\bar{K}_1$ the part corresponding to rotations before the Hamiltonian evolution, $\bar{K}_2$ the part corresponding to rotations after the Hamiltonian evolution, and $\bar{K}_3$ the part corresponding to the Hamiltonian. Let's start by calculating $\bar{K}_1$. Note that

$$\begin{aligned} K_1 &= \sum_{i,l} \frac{\partial \varepsilon}{\partial \theta_{l,i}^1} \frac{\partial \varepsilon}{\partial \theta_{l,i}^1} \\ &= - \langle 0| U_{-,l,1}^\dagger \left[X_l^i, U_{+,l,1}^\dagger \hat{\mathcal{O}} U_{+,l,1} \right] U_{-,l,1} |0\rangle \langle 0| U_{-,l,1}^\dagger \left[X_l^i, U_{+,l,1}^\dagger \hat{\mathcal{O}} U_{+,l,1} \right] U_{-,l,1} |0\rangle . \end{aligned}$$

Let $\rho_0 = |0\rangle \langle 0|$, we have

$$\bar{K}_1 = -\sum_{i,l} \int dU_{-,l,1} \int dU_{+,l,1} \quad \mathrm{Tr} \left(\rho_0 U_{-,l,1}^\dagger P_{+,l,i,1} U_{-,l,1} \rho_0 U_{-,l,1}^\dagger P_{+,l,i,1} U_{-,l,1} \right)$$

$$P_{+,l,i,1} = \left[X_l^i, U_{+,l,1}^\dagger \hat{\mathcal{O}} U_{+,l,1} \right].$$

We now assume that $U_{\pm,l,1}$ are at least 2-designs (i.e. they are approximately Haar random unitaries up to second moment). With this assumption, one can use the results for symbolical integration with respect to the Haar measure on a unitary group [86]. The following calculation follows the very similar calculation done in Appendix B in [25], but there the gates W_l were themselves Haar random. In our case, we have to assume that $U_{\pm,l,1}$ are 2-designs. Let $D = 2^{N+1}$ be the size of the Hilbert space of the QP. Then,

$$\begin{aligned} &\int dU_{-,l,1} \mathrm{Tr} \left(\rho_0 U_{-,l,1}^\dagger P_{+,l,i,1} U_{-,l,1} \rho_0 U_{-,l,1}^\dagger P_{+,l,i,1} U_{-,l,1} \right) \\ &= \frac{\mathrm{Tr}\left(P_{+,l,i,1}\right)^2 \mathrm{Tr}\left(\rho_0^2\right)}{D^2-1} + \frac{\mathrm{Tr}\left(P_{+,l,i,1}^2\right) \mathrm{Tr}\left(\rho_0\right)^2}{D^2-1} + \frac{\mathrm{Tr}\left(P_{+,l,i,1}\right)^2 \mathrm{Tr}\left(\rho_0\right)^2}{D-D^3} + \frac{\mathrm{Tr}\left(P_{+,l,i,1}^2\right) \mathrm{Tr}(\rho_0^2)}{D-D^3} \\ &= \frac{\mathrm{Tr}\left(P_{+,l,i,1}^2\right)}{D^2+D}. \end{aligned}$$

We now average over $U_{+,l,1}$. This requires integrating over $\mathrm{Tr}\left(\left[X_l^i, U_{+,l,1}^\dagger \hat{\mathcal{O}} U_{+,l,1} \right]^2 \right)$. From the identity $\mathrm{Tr}\left([A,B]^2\right) = 2\mathrm{Tr}(ABAB) - 2\mathrm{Tr}(A^2B^2)$, we get the following expression for $\bar{K}_1$:

$$\bar{K}_1 = \frac{2}{D^2+D} \left(\frac{D\mathrm{Tr}(\hat{O}^2) - \mathrm{Tr}(\hat{O})^2}{D^2-1} \right) \mathrm{Tr} \left(\sum_{l,i} \left(X_l^i\right)^2 \right) = \frac{2L(N+1)}{D^2+D} \left(\frac{D\mathrm{Tr}(\hat{O}^2) - \mathrm{Tr}(\hat{O})^2}{D^2-1} \right). \tag{C10}$$

A similar calculation can be done for $\bar{K}_2$ yielding the result $\bar{K}_2 = \bar{K}_1$. Now, for the case of $\bar{K}_3$ note that

$$\bar{K}_H = -\sum_l \int dU_{-,l} \int dU_{+,l} \quad \mathrm{Tr} \left(\rho_0 U_{-,l}^\dagger P_{+,l} U_{-,l} \rho_0 U_{-,l}^\dagger P_{+,l} U_{-,l} \right)$$

$$P_{+,l} = \left[A_i, U_{+l}^\dagger \hat{\mathcal{O}} U_{+,l} \right].$$

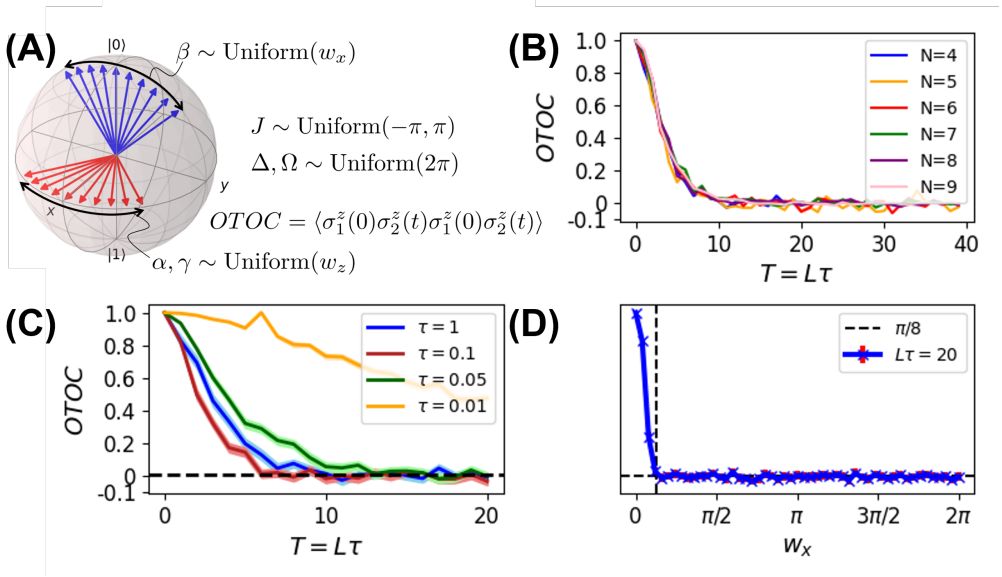

FIG. 9. Results of measuring OTOCs on QPs. (A) A schematic of the ensemble of QPs used for the other panels. The couplings, fields, and rotation angles are all sampled uniformly from different intervals. The OTOC measured is $\langle \sigma_1^z(0)\sigma_N^z(T)\sigma_1^z(0)\sigma_N^z(T)\rangle$ where $T = L\tau$ is the total time. (B) OTOCs for different values of N showing the behavior of the OTOC. The convergence to a unitary 2-design is independent of N. (C) OTOCs for $N = 3$ and different values of τ. For $\tau > 0.01$, the total time T largely characterizes the layers needed to converge to the Haar prediction signaling that the QP can approximate a unitary 2-design. For small τ, a much larger total time $T \sim 1/(\langle J^2\rangle\tau)$ is needed (see Fig. 10). (D) Dependence of the OTOC on w_x, the scale of the single-qubit rotations around the x-axis. For this panel $N = 3$, $L = 20$, and $\tau = 1$. For $w_x > \pi/8$, a QP approximates a unitary 2-design.

Thus, we can apply the same mathematical procedure to this part of the QNTK if we assume $U_{\pm,l}$ are also 2-designs. Since A_i^2 is the identity, we have

$$\bar{K}_3 = \frac{2}{D^2 + D}\left(\frac{D\mathrm{Tr}(\hat{O}^2) - \mathrm{Tr}(\hat{O})^2}{D^2 - 1}\right)\mathrm{Tr}\left(\sum_{l,i}(A_i)^2\right) = \bar{K}_1.$$

Thus, we arrive at the result that

$$\bar{K} = \frac{6L(N+1)}{D^2 + D}\left(\frac{D\mathrm{Tr}(\hat{O}^2) - \mathrm{Tr}(\hat{O})^2}{D^2 - 1}\right) \tag{C11}$$

1. QPs and 2-designs

The assumption that a VQA forms a unitary 2-design is common in the QML community when looking at th QNTK. For example, in the variational algorithms treated in [24, 25, 87] every qubit is coupled to each other via two-qubit gates, which are architectures that can be mapped to a Clifford circuit. As a result of this mapping, the algorithm can approximate a 2-design when $L \sim \mathcal{O}((N+1)^2)$ [87]. In the case of a QP, however, the connectivity is limited. The inputs do not couple to each other in a single layer of the QP.

One way to measure randomness in a quantum circuit is by looking at the so-called "out-of-time-order correlator" (OTOC) [32]. In our case, we work with the OTOC

$$\mathrm{OTOC}(T) = \langle \sigma_1^z(0)\sigma_N^z(T)\sigma_1^z(0)\sigma_N^z(T)\rangle \tag{C12}$$

where $\sigma_i^z(T) = U\sigma_i^z U^\dagger$ with U being the QP unitary for a total time $T = L\tau$. If the unitary U were Haar random, then for long times T, we would obtain $OTOC \to -1/(D^2 - 1)$. Note that Eq. (C12) is related to the commutator of the two observables via $\mathrm{Re}(OTOC(T)) = 1 - \frac{1}{2}\langle[\sigma_1^z(T), \sigma_2^z]^2\rangle$. Thus, the OTOC is a measure of the non-commutativity of two evolved operators which were originally local. Moreover, if an OTOC approximates $-1/(D^2 - 1)$ for an ensemble

of circuits, then the ensemble approximates a 2-design [31]. As a result, measuring the average OTOC for an ensemble of QPs tells us about the validity of assuming they are at least approximately unitary 2-designs.

Fig. 9 shows the results of measuring the OTOC on a QP. Fig. 9(A) is a schematic of the ensemble of QPs used in all other panels. The couplings J are sampled uniformly from the interval $[-\pi, \pi]$, while the fields Δ_o^l and Ω_o^l are sampled uniformly from the interval $[0, 2\pi]$. The single-qubit rotations are sampled uniformly in the intervals $[0, w_x]$ and $[0, w_z]$ respectively. For panels (B) and (C), $w_x = w_z = 2\pi$. We note that Eq. (C12) consists of operators on two input nodes and thus quantifies the amount of scrambling on the input qubits due to the interaction with the output qubit in a QP.

Fig. 9(B) shows the behavior of the OTOC for several values of N. Note that here $\tau = 1$. The OTOC converges to the 2-design prediction at a constant depth $L \sim 10$ independent of N. One way of explaining this is that the QPs connectivity renders the OTOC independent of N since any two input qubits connect via only one intermediate output qubit. That is, any initially local pair of operators is separated by at most two sites regardless of the total size of the QP.

Fig. 9(C) shows the OTOC for $N = 3$ and for different values of τ. For $\tau > 0.01$, the results are similar as long as the total time $T = L\tau$ is the same. However, for $\tau < 0.01$, the OTOC fails to achieve the Haar value even for depths $L = 1000$. This discrepancy comes from the fact that this OTOC measures the operator spreading of operators initially living on two inputs. We can model the operators spread by the rate at which the qubits become entangled. However, the entanglement of two inputs is a second-order effect as evidenced in Eq. (B4). For L layers, two inputs become entangled at an average rate $r = \frac{\langle J^2\rangle \tau^2 L}{2}$ which is of order unity if $T \sim 1/(\tau\langle J^2\rangle)$. To show this, note that the unitary operation in question (for small τ) can be written as

$$U \approx 1 - i\tau \sum_i J_i \sigma_i^z \sigma_o^z - \frac{\tau^2}{2} \sum_{ik} J_i J_k \sigma_i^z \sigma_k^z \tag{C13}$$

where we have ignored the local terms on the output. As it stands, U is diagonal in the z-basis and thus cannot operate on operators such as σ_i^z nontrivially.

However, the local rotations effectuate a nontrivial Hamiltonian since the Pauli-matrices get rotated in average to a new operator $\bar{\sigma}_i$ depending on the choices of w_x and w_z (as a sanity check, w_x returns the original Pauli matrices). So we can rewrite the approximation after averaging out the local rotations as

$$U \approx 1 - i\tau \sum_i J_i \bar{\sigma}_i \bar{\sigma}_o^z - \frac{\tau^2}{2} \sum_{ik} J_i J_k \bar{\sigma}_i \bar{\sigma}_k^z \tag{C14}$$

Now, we can answer how fast this Hamiltonian takes to spread two operators initially on two input qubits. For $w_x \neq 0$, the rate at which this happens is $\tau^2 J_i J_j/2$. Taking an average over the values of J_i, each layer of a QP spreads the operators at the rate $\frac{\langle J^2\rangle\tau^2}{2}$. For L layers, the rate is then

$$r = \frac{\langle J^2\rangle \tau^2 L}{2} \tag{C15}$$

if we want this to be of order $\mathcal{O}(1)$ such that substantial spreading occurs, then we need $T = 2/(\langle J^2\rangle\tau)$.

Suppose instead that we wanted to look at the OTOC

$$\langle \sigma_o^z(t) \sigma_N^z \sigma_o^z(t) \sigma_N^z \rangle \tag{C16}$$

between an input and the output neuron. In this case, the operator spreads at the rate $r = L\tau\langle J\rangle$ which is of order unity when $T \sim 1/\langle J\rangle$. Since in our case $\langle J\rangle = \pi$, $T \sim 1$ is when the OTOC starts decaying to close to zero. Indeed, in Fig. 10 shows that for the OTOC in Eq. (C16) behaves similarly for different values of τ as long as the total time $L = \tau L$ is consistent. This is in sharp contrast with Fig. 9(C) where for small τ the time T is insufficient to characterize whether the OTOC is decaying. This is due to the fact that operator spreading of operators initially in the output happens as a second order effect at the rate r in Eq. (C15).

Lastly, note that if $w_x = 0$ then no operator spreading would be possible. This is consistent with what we see in Fig. 9(D). In fact, the numerics in Fig. 9(D) show that, as long as $w_x > \pi/8$, the effective Hamiltonian in Eq. (C14) is sufficiently entangling when $r \sim 1$.

Finally, Fig. 9(D) shows the dependence of the OTOC on w_x. For this panel, $N = 3$, $\tau = 1$ and $L = 20$. The average OTOC is calculated for increasing values of w_x and shows that when $w_x > \pi/8$, the OTOC follows the Haar prediction. The importance of $\pi/8$ is that rotation gates of at least that angle allow one to approximate arbitrary single-qubit rotations [23]. The width w_z does not affect the OTOC. Note that $w_z > \pi/8$ alone does not allow for

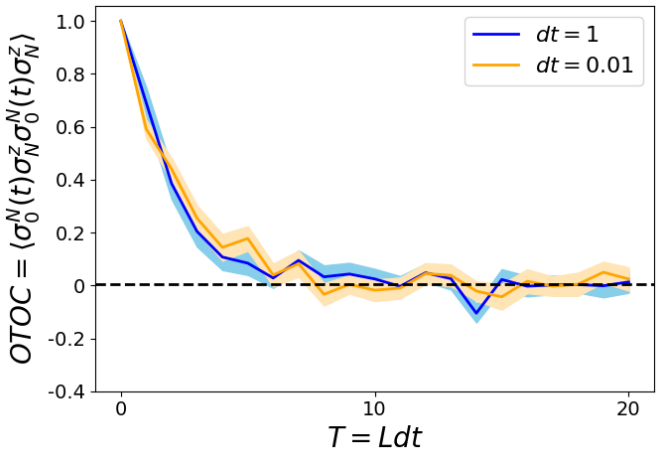

FIG. 10. OTOCs on an input qubit and the output qubit as a function of $T = \tau L$ for two different values of $\tau = 1, 0.01$. Note that the total time T suffices to characterize the time at which the OTOC starts decaying towards zero ($T \sim 1$).

nontrivial dynamics since, although we have T-gates, we cannot approximate Hadamard gates readily. But when w_x is also greater than $\pi/8$, $R_{x,z}(\pi/4)$ readily approximate the Hadamard- and T- gates needed for universal quantum computation since Hadamard is proportional to $R_x(\pi/2)R_z(\pi/2)R_x(\pi/2)$. Thus, the width w_x serves as a knob that enhances a QP from performing only nontrivial operators on the output node ($w_x = 0$) to being able to do nontrivial operations on the input when ($w_x > \pi/8$).

The results in this section support the assumption that QPs can approximate 2-designs. This assumption led to the QNTK results in Eq. (26).

Appendix D: Details on the numerical implementations of QPs

While in the paper all of our numerics are for system sizes $N \leq 10$, we remark that scalable simulations of QPs are available. To see this, it is useful to look at how our QPs fit into a matrix product state framework (MPS) [33, 34]. An MPS represents a quantum state as the contraction of tensors

$$|\psi\rangle = \sum_{\boldsymbol{s}} \left(A^{s_1}_{i_1} A^{s_2}_{i_1 i_2} \dots A^{s_N}_{i_{N-1}} \right) |s_1 s_2 \dots s_N\rangle \tag{D1}$$

where $\boldsymbol{s}$ is a binary vector of length N, $A^{s_1}_{i_1}$ and $A^{s_N}_{i_N}$ are vectors of size $d \times 1$ and $1 \times d$ respectively, and matrices $A^{s_k}_{i_k}$ for $1 < k < N$ are matrices of size $d \times d$. The parameter d is called bond-dimension, and it physically corresponds to the degree of entanglement shared between qubits k and $k \pm 1$. It is a convention to choose D to be uniform for all matrices. We can denote $A^{[k]}$ as the 3-tensor of dimension $2 \times d \times d$ containing information about qubit k. Note that these tensors have a dimension of size 2 which corresponds to the physical index (or leg) of the qubit, and two dimensions of size d which correspond to virtual legs to neighboring qubits. These virtual legs contain information about the entanglement structure of the state. While a state $|\psi\rangle$ would normally be described by the coefficients $\psi(\boldsymbol{s})$ for each basis state, the matrices in Eq. (D1) can be found from these coefficients via Smidth-decomposition [88].

Using similar logic, operators can also be written in terms of matrix products leading to the notion of a matrix product operator (MPO)

$$\hat{\mathcal{O}} = \sum_{\boldsymbol{s},\boldsymbol{c}} M^{s_1,c_1}_{i_1} M^{s_2,c_2}_{i_1,i_2} \dots M^{s_N,c_N}_{i_{N-1}} |s_1 s_2 \dots s_N\rangle\langle c_1 c_2 \dots c_N|. \tag{D2}$$

The tensors $M^{[k]}$ are now $2 \times 2 \times d_{\text{op}} \times d_{\text{op}}$ dimensional (with a bond dimension d_o possibly different than that of an MPS) containing two physical indexes and two virtual indexes. The virtual indexes contain information as to how the operator entangles neighboring qubits. MPOs can be applied to an MPS producing a new MPS with new bond-dimension dd_{op}. The process of propagating an MPS through a series of L operators takes run-time $\mathcal{O}(Nd(d_{\text{op}})^L)$ [33]. This run time is smaller than the full-state description as long as $L < N$ and d is small.

Let us now apply the MPS framework to our QPs. First of all, single-qubit rotations R_k can be mapped into MPOs with $d_{\text{op}} = 1$ where $M^{s_k,c_k} = \langle c_k|R_k|s_k\rangle$ is the matrix element of the rotation (a scalar). Secondly, and more

interestingly, evolving under Eq. (4) can also be done via first mapping the Hamiltonian into an MPO. Here we must remark that not all Hamiltonians have exact MPO representations. Moreover, H_P is, in some sense, a long-range Hamiltonian since qubit $i = 1$ interacts with the output which in the MPS language is N sites away. However, this apparent long-range is due to the linear geometry of the MPS form, and physically all input qubits are neighbors to the output. This realization is what allows us to derive the exact MPO form

$$H_P^{[1]} = \begin{pmatrix} \mathbb{1}_1 & J_1\sigma_1^z \end{pmatrix}, \quad H_P^{[k]} = \begin{pmatrix} \mathbb{1}_k & J_k\sigma_k^z \\ 0 & \mathbb{1}_k \end{pmatrix}, \quad H_P^{[o]} = \begin{pmatrix} \Omega_o\sigma_o^x - \Delta_o\sigma_o^z \\ \sigma_o^z \end{pmatrix}. \tag{D3}$$

Note that with this notation $H_P = H^{[1]}H^{[2]}\dots H^{[o]}$. The bond dimension of H_P is $d_{\text{op}} = 2$. It has been shown that for a Hamiltonian with an MPO, its corresponding unitary evolution, $U = e^{-i\tau H_P}$ in our case, can be approximated by an MPO as well [34]. In particular, since we cant to consider the effect of all fist and second order terms in τ, we use the approximation

$$U \approx= 1 - i\tau H_P - \frac{\tau^2}{2}H_P^2 \tag{D4}$$

the multiplication of two MPOs of bond dimension d_{op} results in an MPO of bond dimension d_{op}^2, while addition results in adding the bond dimensions. So our approximation has a bond dimension $1 + d_{\text{op}} + d_{\text{op}}^2 \sim \mathcal{O}(d_{\text{op}}^2)$. Thus, the algorithm scales as $\mathcal{O}(Nd_{\text{op}}^{2L})$ which is efficient for $L < N/2$. The error in this approximation scales as $\mathcal{O}(L\tau^3)$. If $\tau \ll 1$ one could also use a first degree approximation of U with an error scaling as $\mathcal{O}(L\tau^2)$ but with a bond dimension of order d_{op} such that the algorithms is efficient for $L < N$.

Appendix E: Supplemantary calculations on using QPs for quantum metrology

In this section, we explain how to measure the signal of Sec. V D using the output qubit. Suppose that after the circuit in Fig. 6, and before the measurement, the QP's state is

$$\sum_{\boldsymbol{y}} \psi(\boldsymbol{y})|\boldsymbol{y}\rangle|-\rangle \tag{E1}$$

where $|\boldsymbol{y}\rangle = |y_1, y_2, ..., y_N\rangle$ is a separable state where $\sigma_i^y|y_i\rangle = y_i|y_i\rangle$. Now, suppose that we first rotate all the input qubits by $R_x(\pi/2)$ resulting in $\sigma_i^z \to \sigma_i^y$. Then using an H_P layer with $\Omega_o = \Delta_o = 0$ and $J_i = J$ we obtain

$$\sum_{\boldsymbol{y}} \psi(\boldsymbol{y})|\boldsymbol{y}\rangle|-\rangle \to \sum_{\boldsymbol{y}} \psi(\boldsymbol{y})|\boldsymbol{y}\rangle e^{-iJ\tau\sum_i y_i\sigma_o^z}|-\rangle_o \tag{E2}$$

Noticing that $\sum_i y_i = 2\langle\boldsymbol{y}|S_y|\boldsymbol{y}\rangle$ we see that the probability of observing the output neuron in the state $|+\rangle_o$ is

$$P_+ = \sum_{\boldsymbol{y}} \sin^2(2Jt\langle\boldsymbol{y}|S_y|\boldsymbol{y}\rangle)|\psi(\boldsymbol{y})|^2. \tag{E3}$$

For small $NJt \ll 1$ we obtain to second order in t

$$P_+ = (2Jt)^2\langle S_y\rangle^2 = (2Jt)^2(m\varphi N/2)^2. \tag{E4}$$

Therefore, with a circuit performing a global rotation on the input and a small-time evolution under H_P, we can use the output qubit to measure the signal φ and the amplification m.